Machine Learning

Boris Velichkov

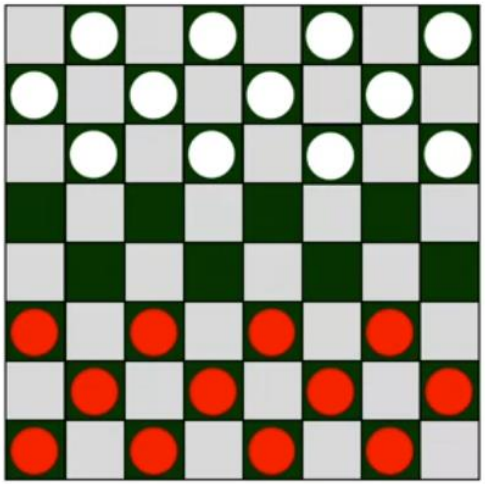
Machine Learning?



Machine Learning

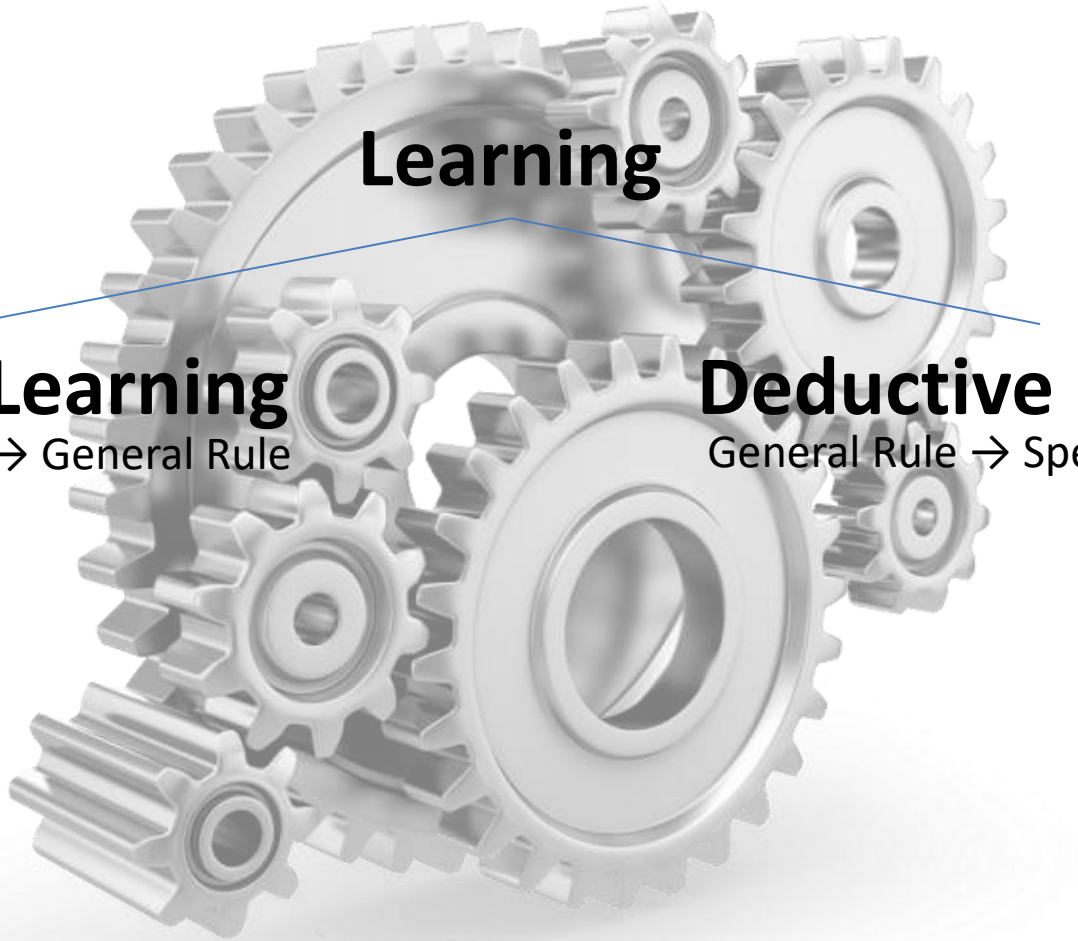
"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



Checkers

How do we learn?



Learning

Inductive Learning

Specific Examples → General Rule

Deductive Learning

General Rule → Specific Examples

Machine Learning *is …*

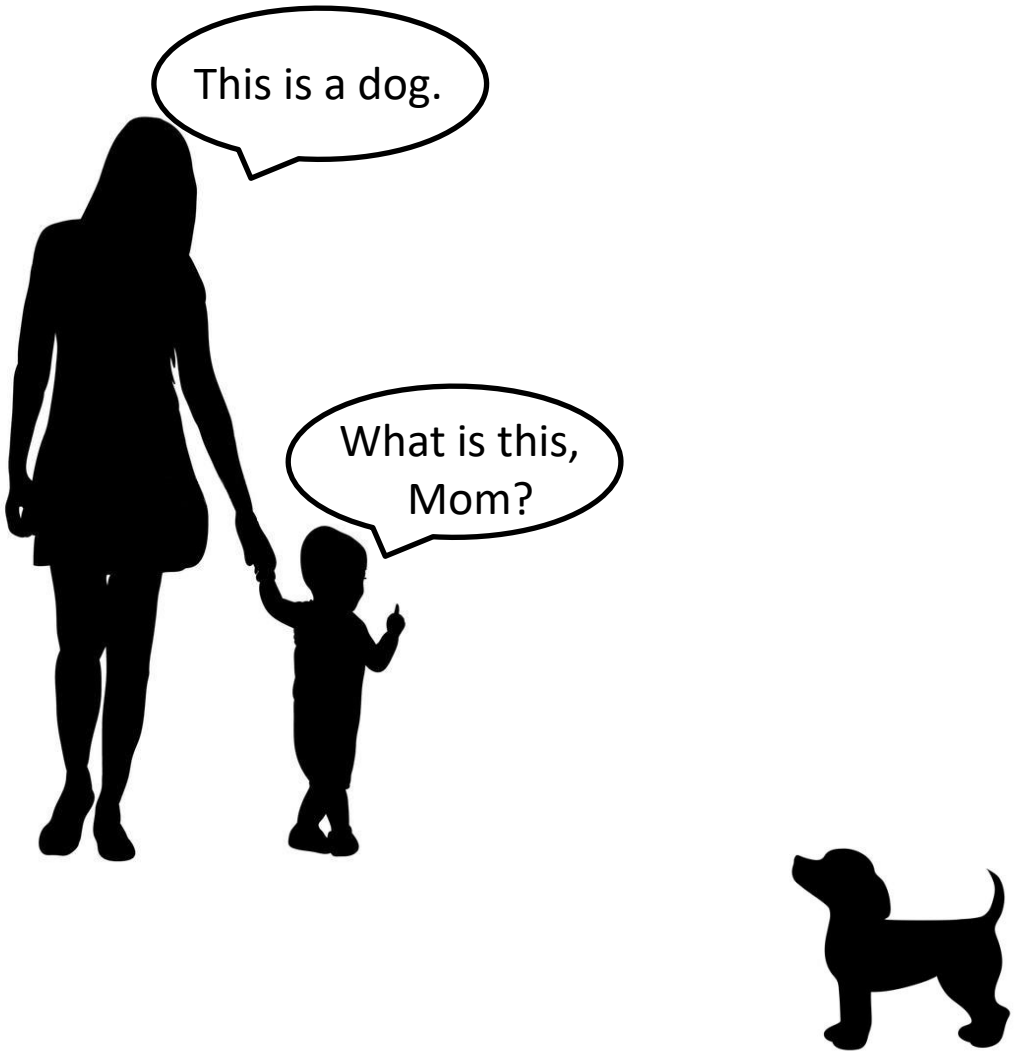
Machine Learning *is* Inductive Learning

How do we learn? Example?

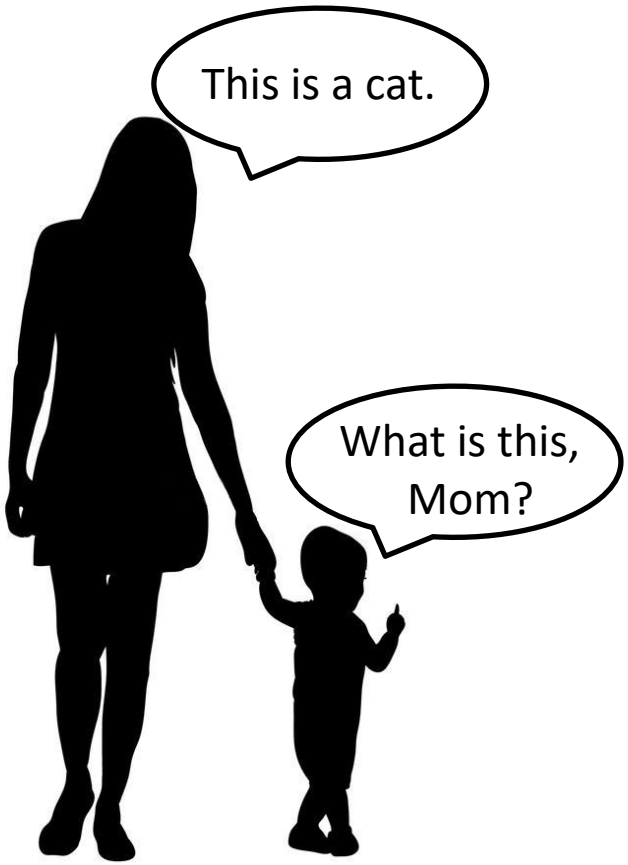


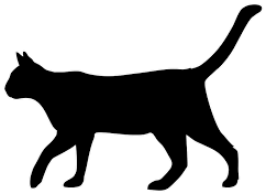


How do we learn – Example 1

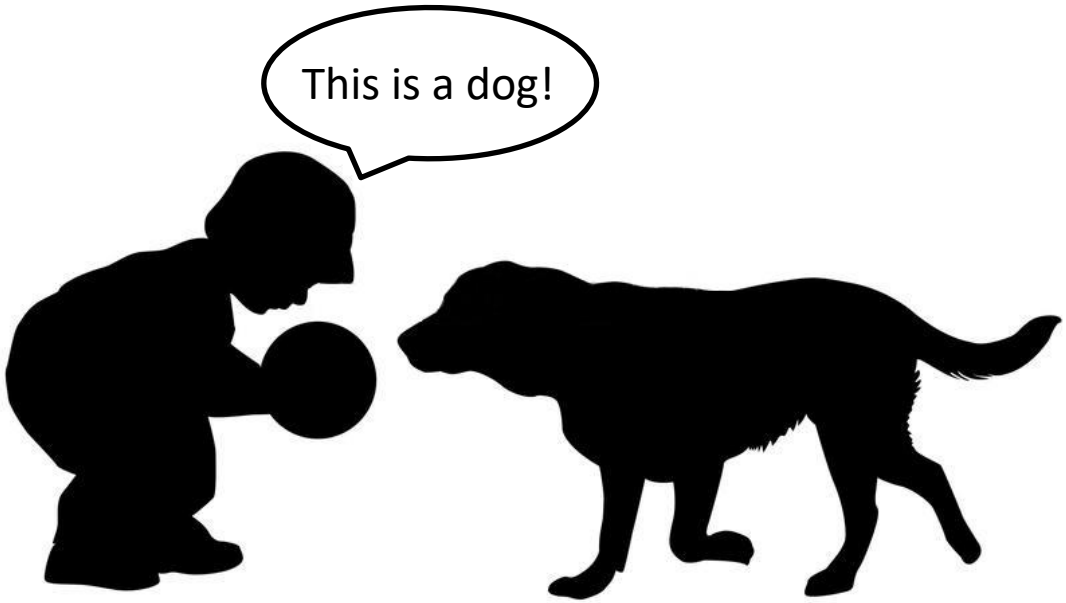


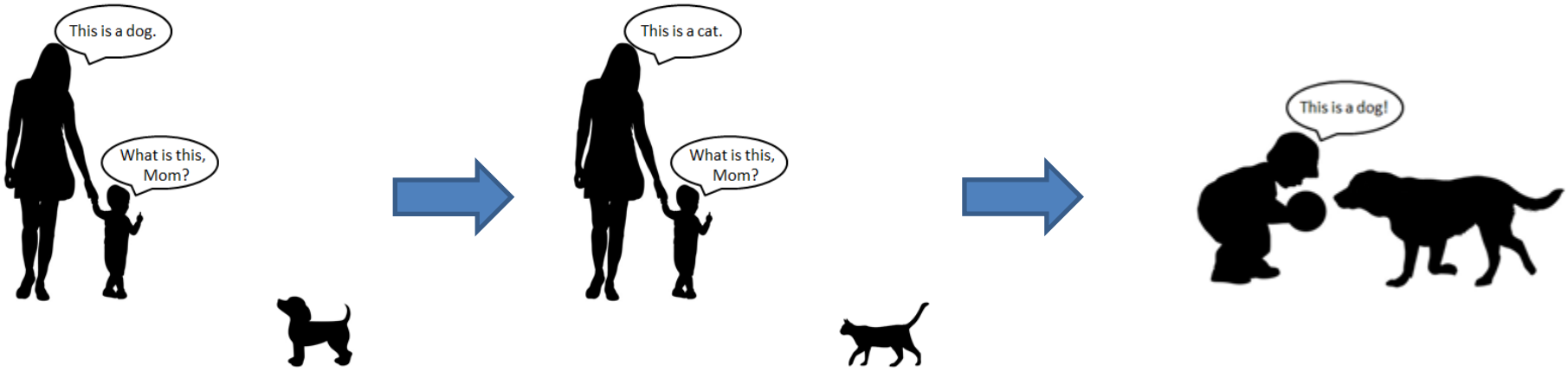
How do we learn – Example 1



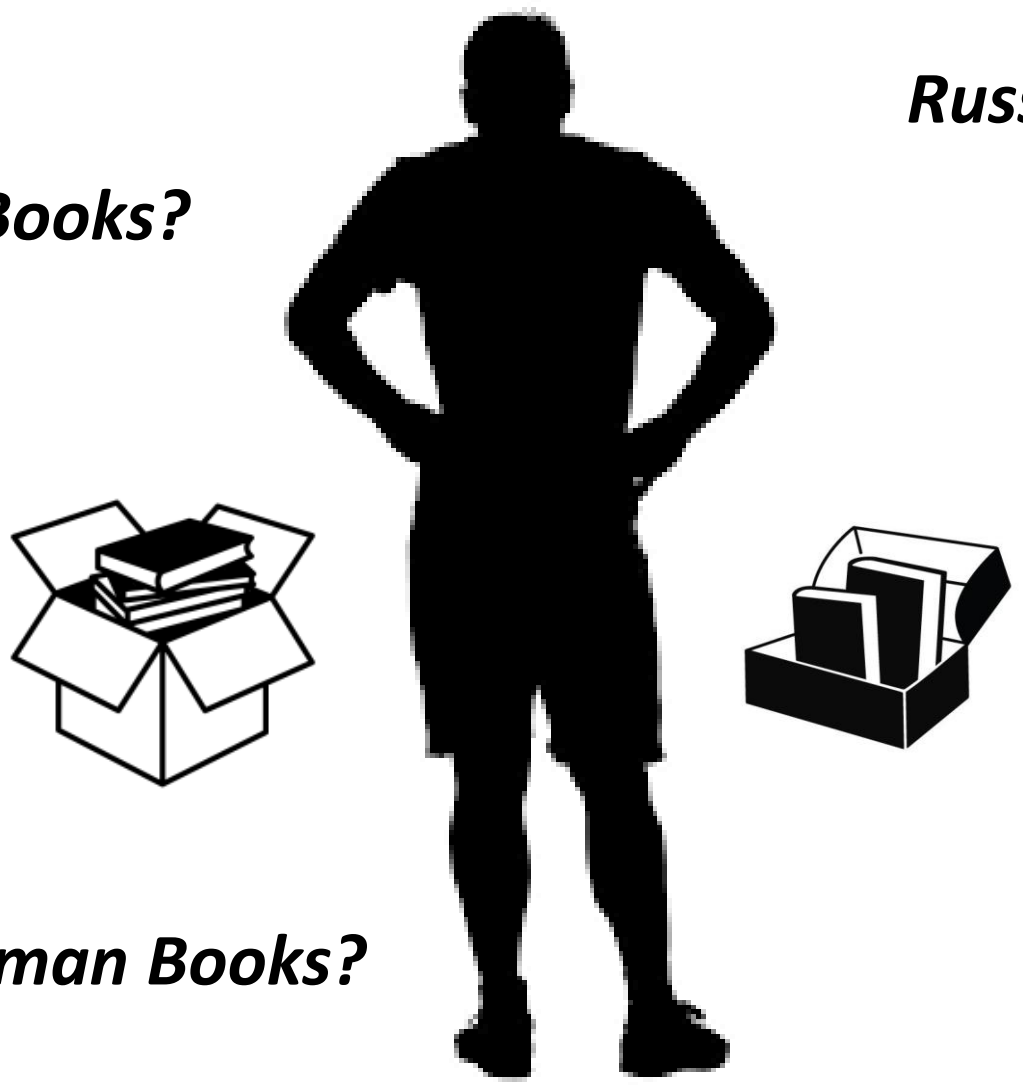


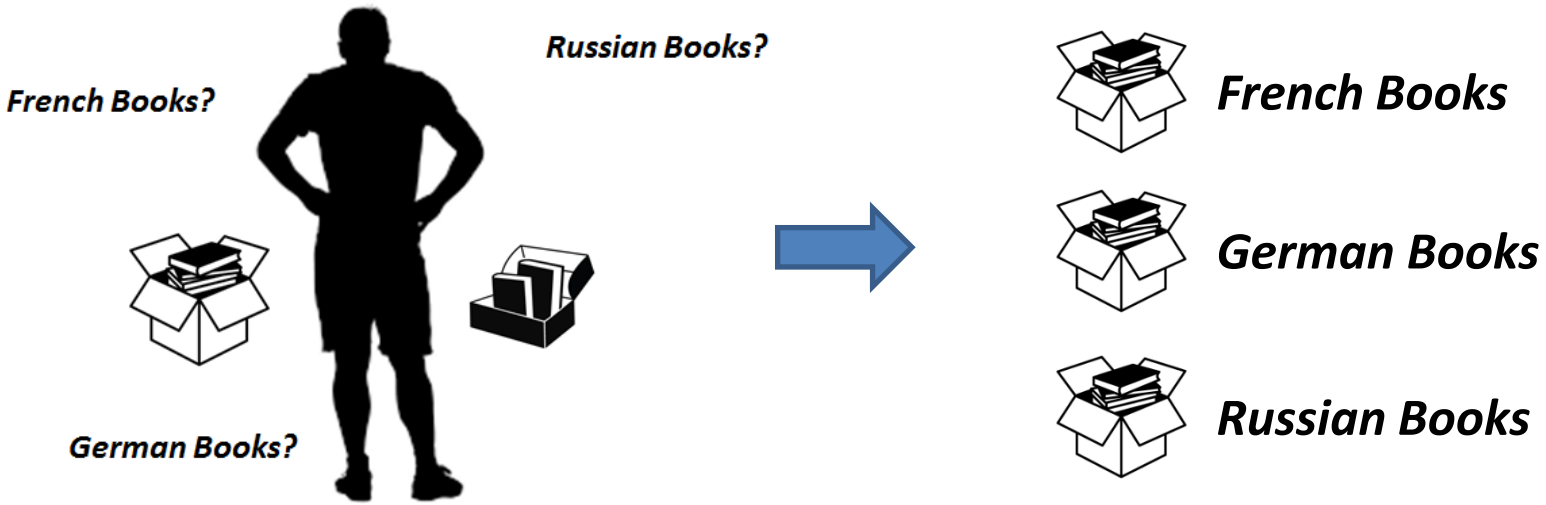
How do we learn – Example 1

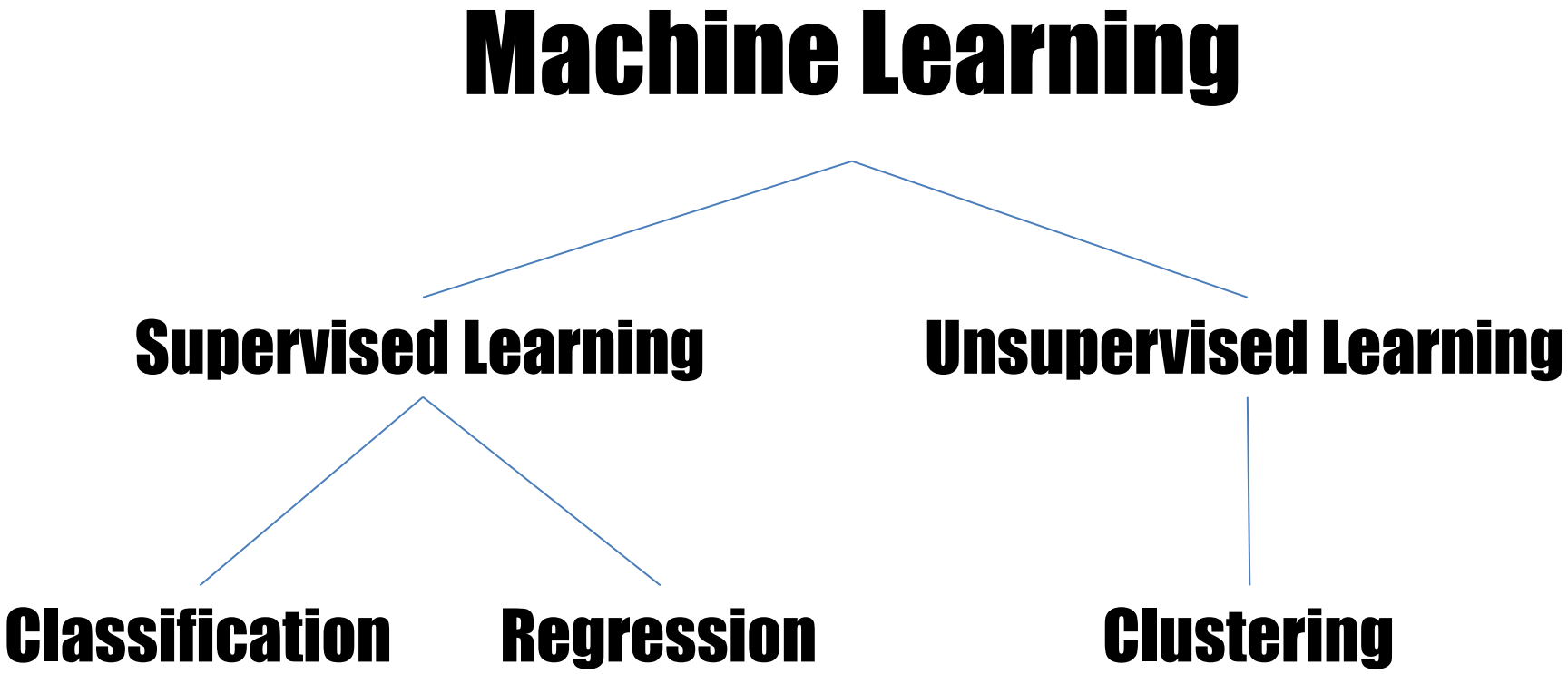



This is a dog.
What is this, Mom?
This is a cat.
What is this, Mom?
This is a dog!

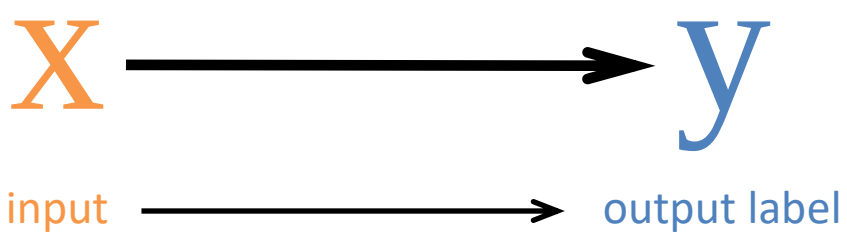
How do we learn – Example 2




French Books?
Russian Books?
German Books?
French Books
German Books
Russian Books


Machine Learning
Supervised Learning
Unsupervised Learning
Classification
Regression
Clustering

Supervised Learning

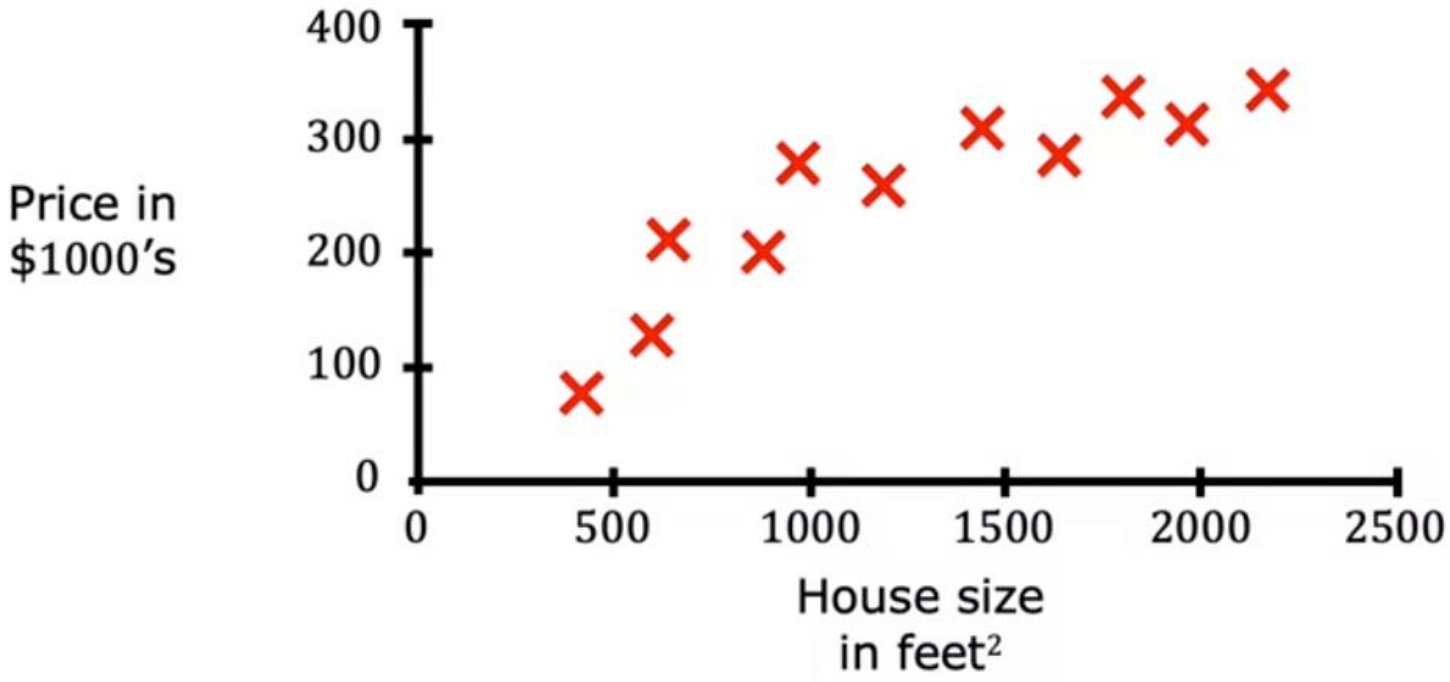


Supervised Learning

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car
image of phone	→	defect? (0/1)	visual inspection

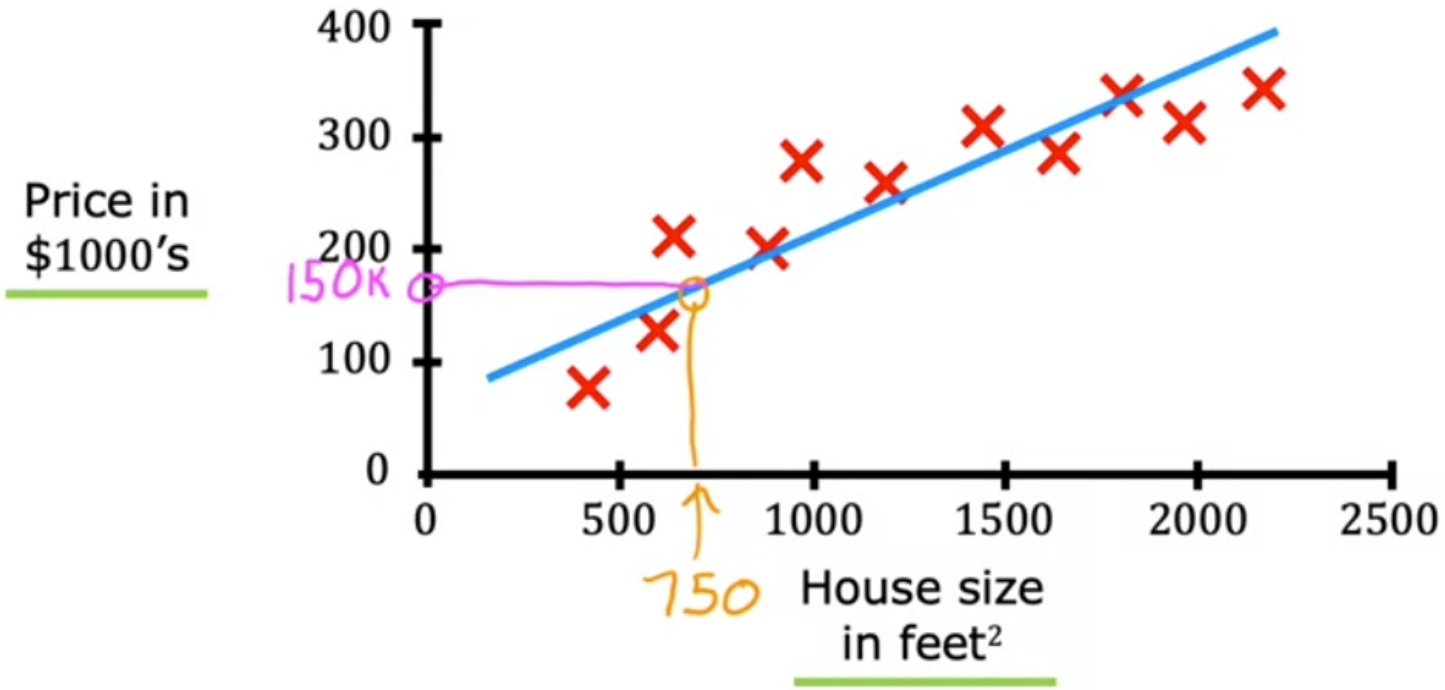
Supervised Learning

Regression: Housing price prediction



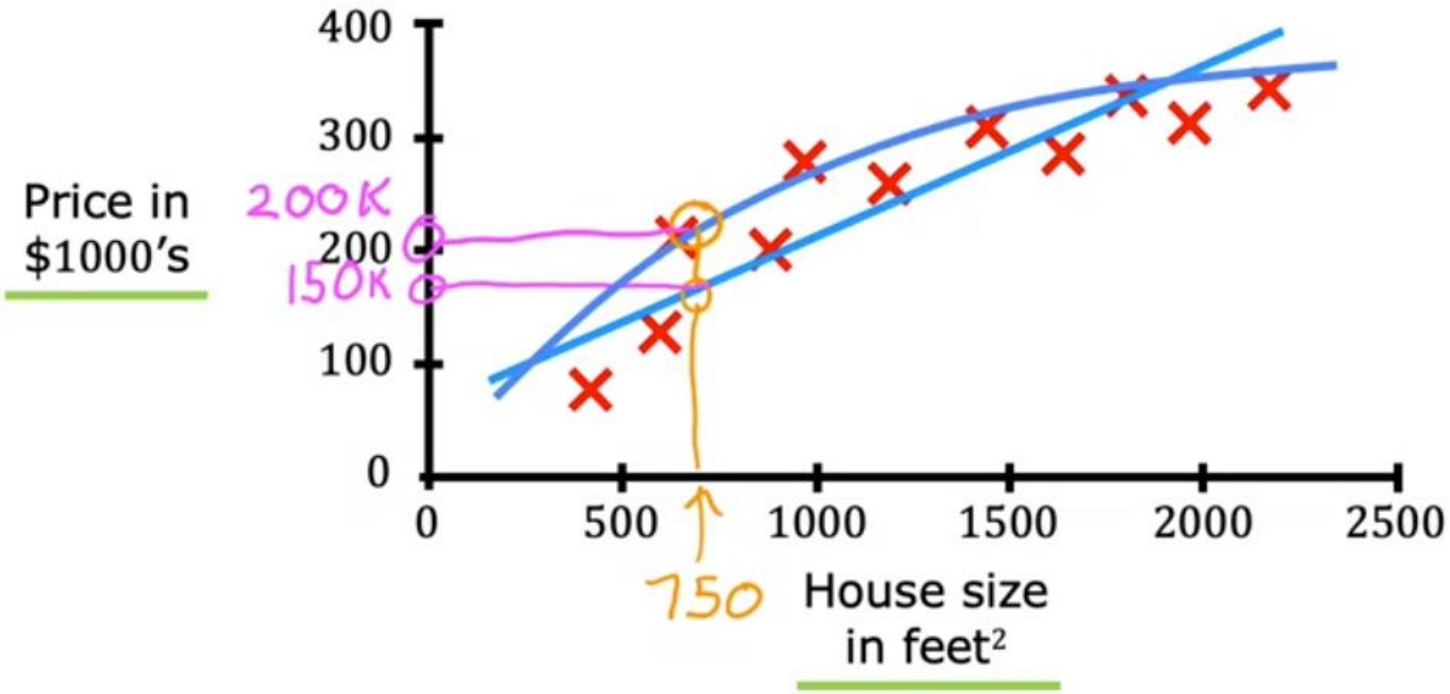
Supervised Learning

Regression: Housing price prediction

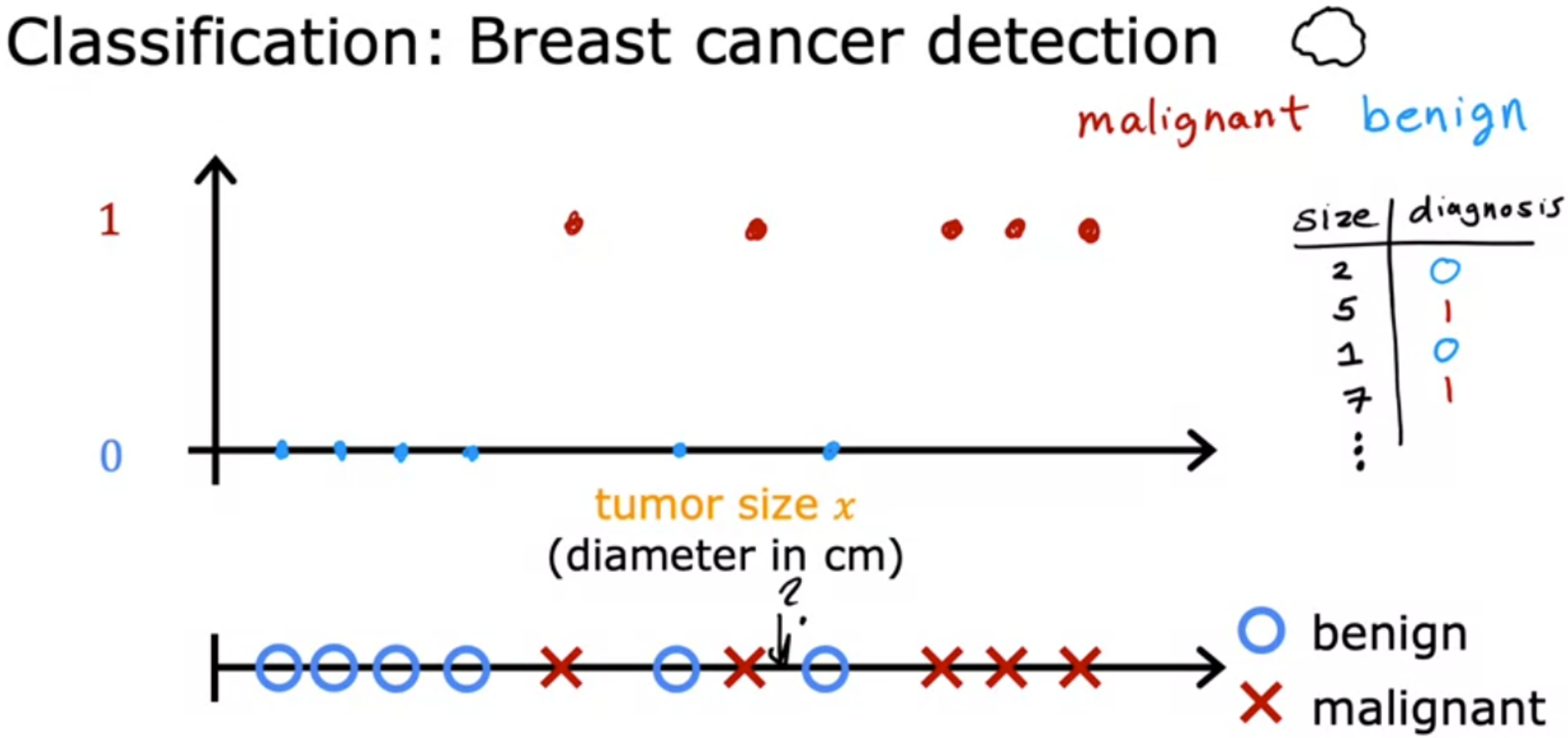


Supervised Learning

Regression: Housing price prediction

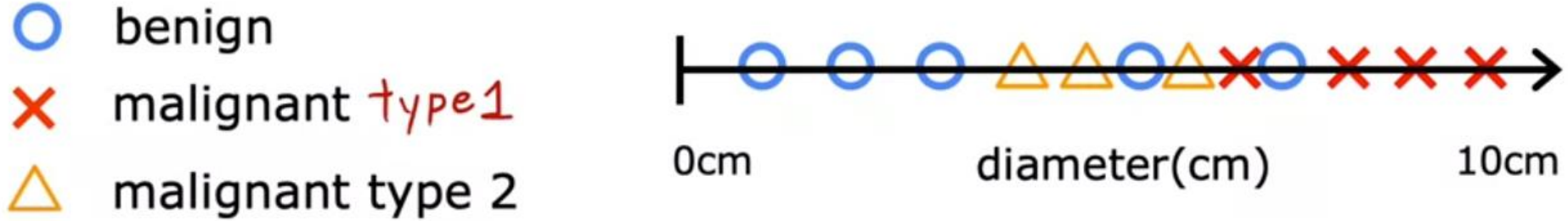


Supervised Learning



Supervised Learning

Classification: Breast cancer detection



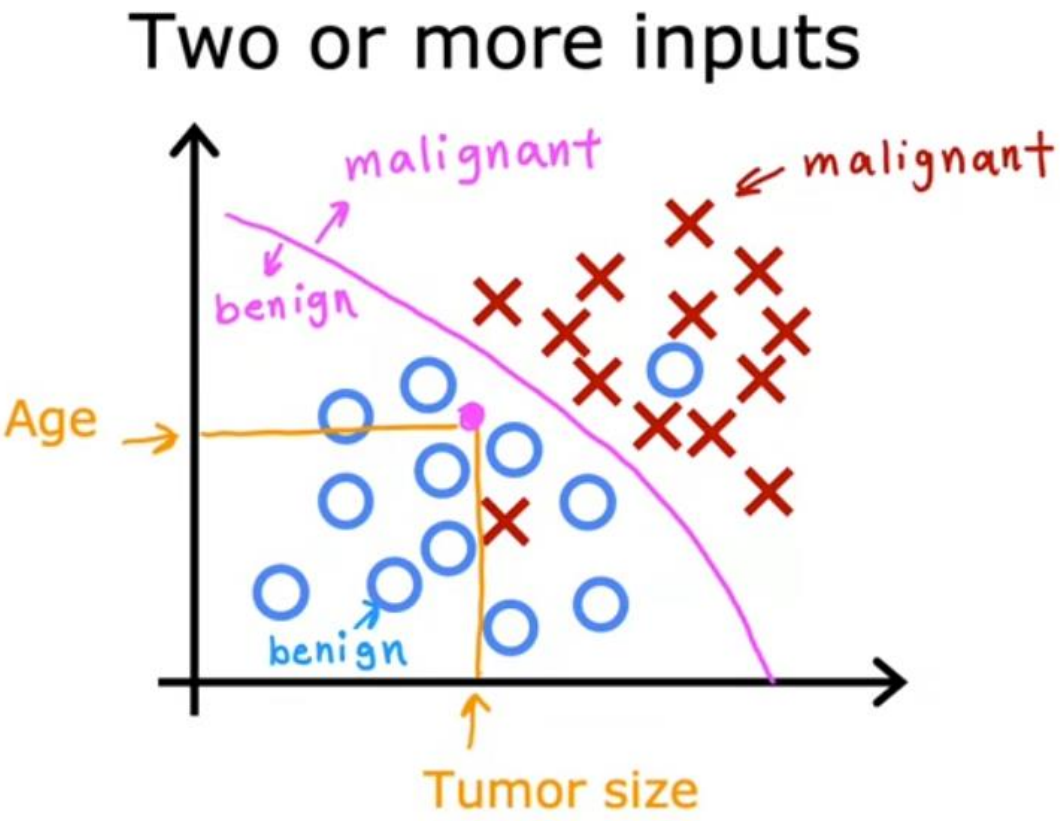
class category

Classification

predict categories cat dog benign malignant 0, 1, 2

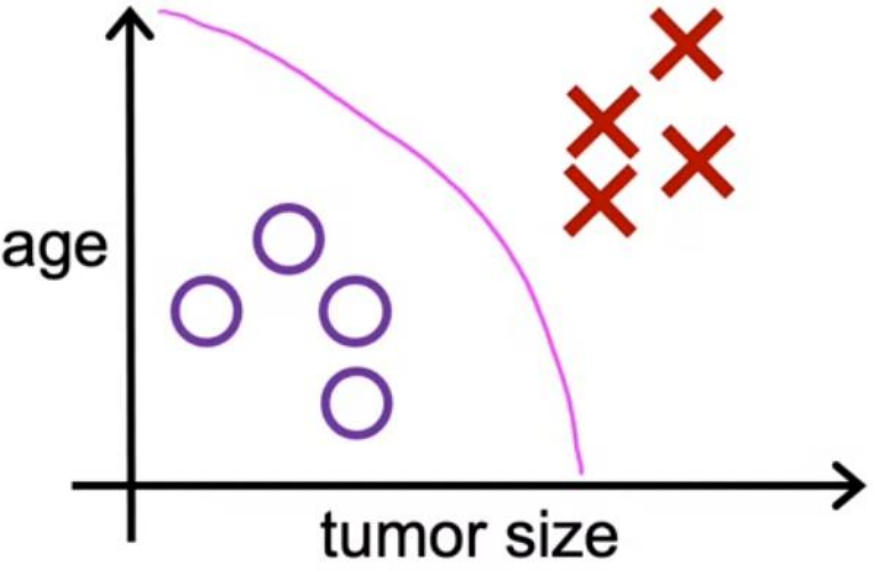
small number of possible outputs

Supervised Learning

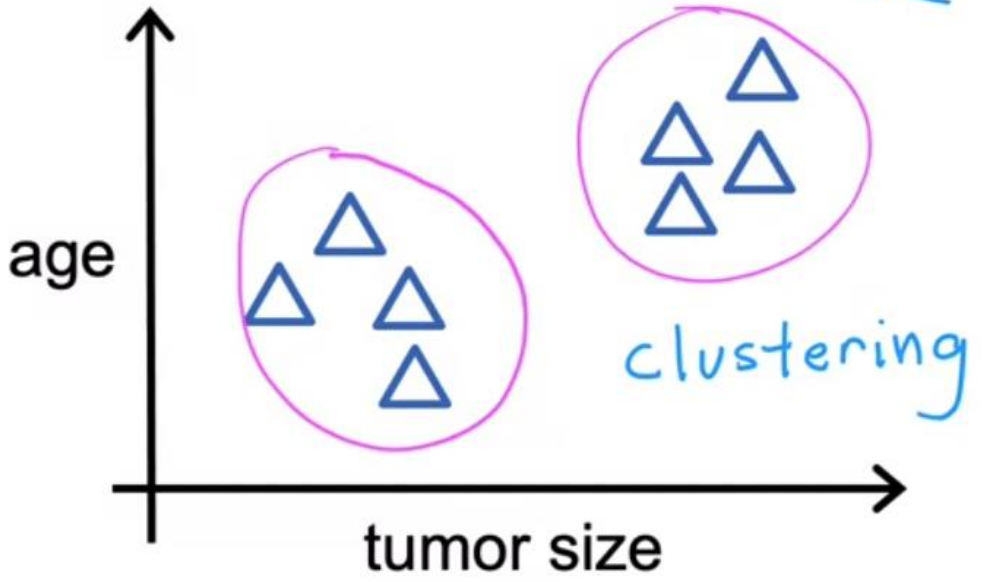


Supervised vs Unsupervised Learning

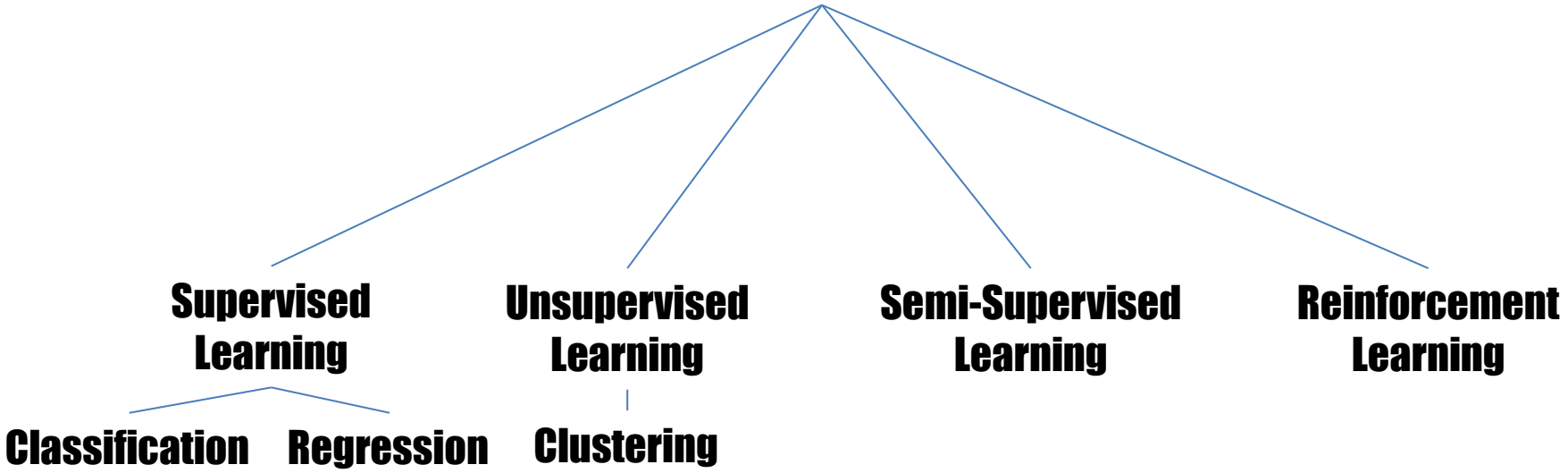
Supervised learning
Learn from data labeled with the "right answers"



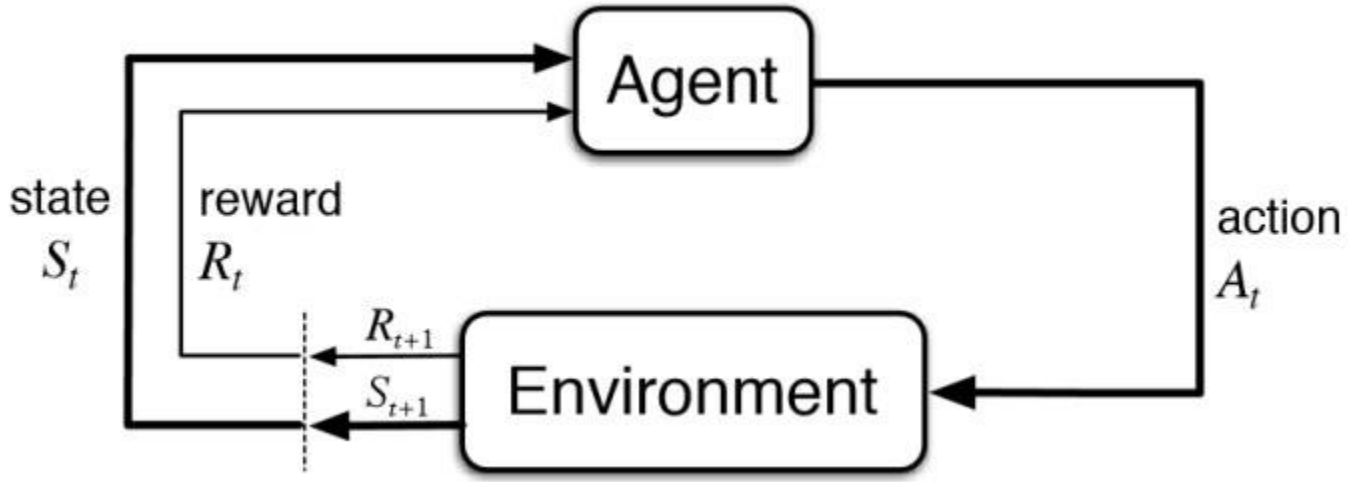
Unsupervised learning
Find something interesting in unlabeled data.



Machine Learning

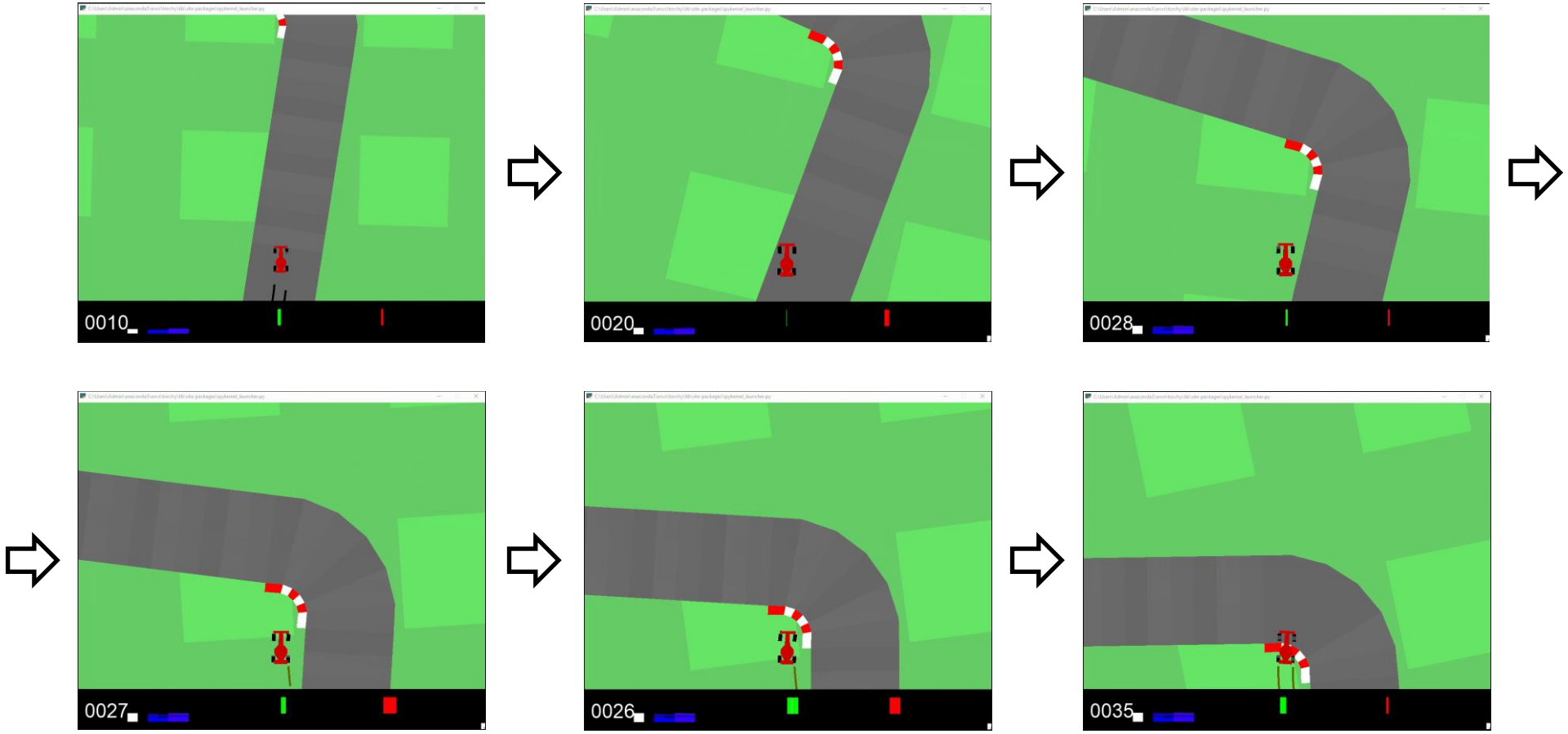


Reinforcement Learning



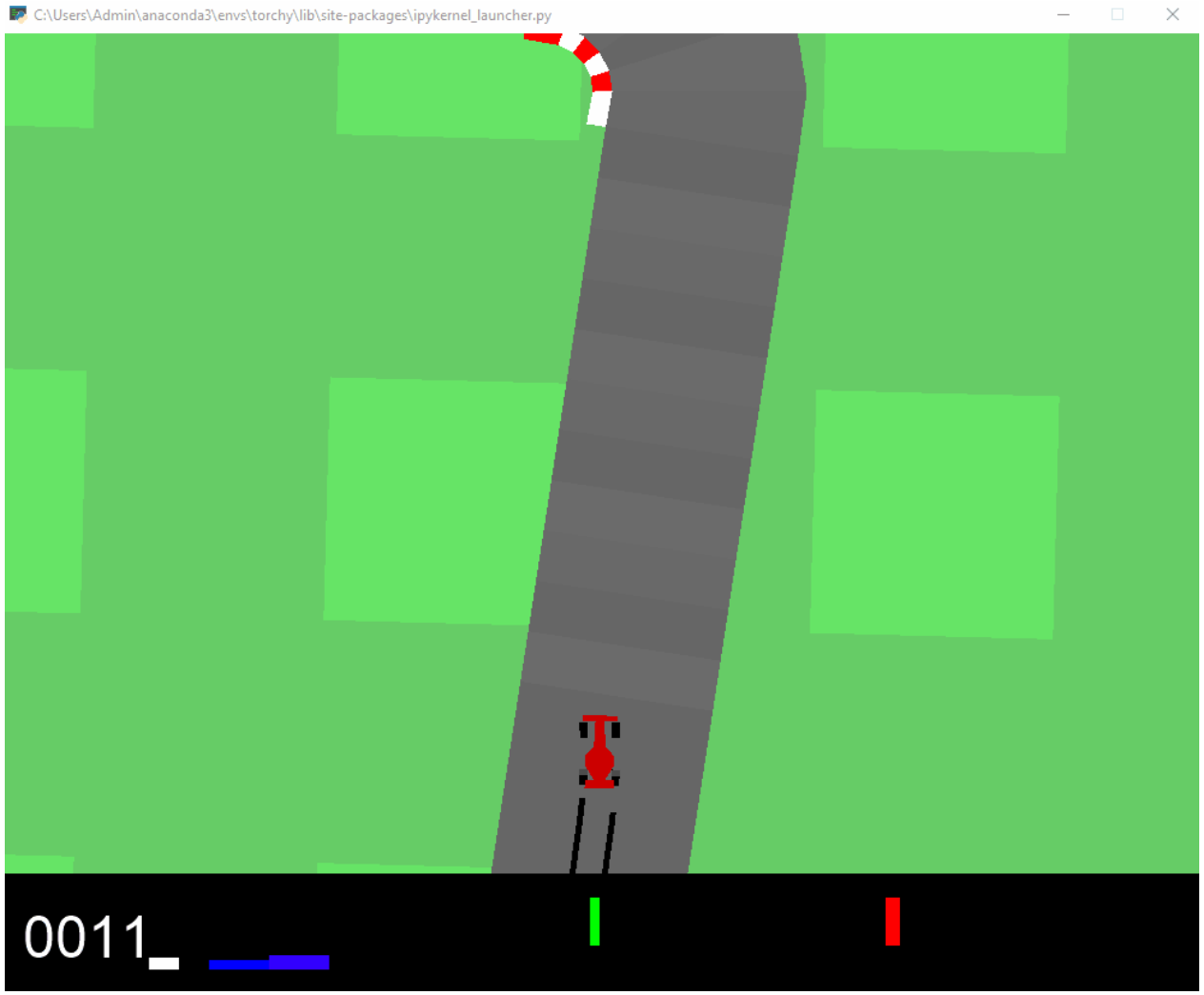
Reinforcement Learning

Example #1



Reinforcement Learning

Example #1



Reinforcement Learning

Example #2



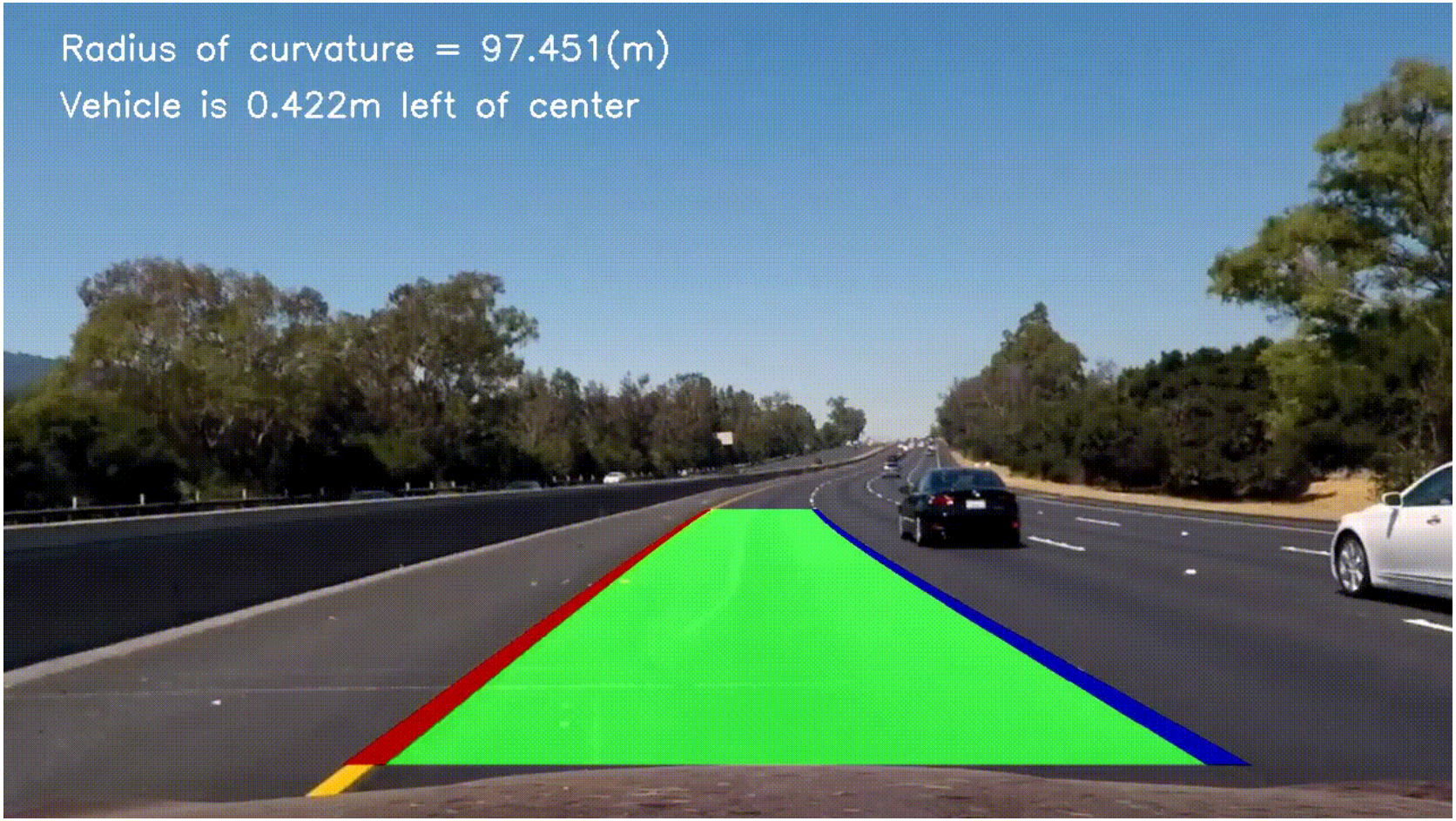
Reinforcement Learning

Example #2



Reinforcement Learning

Example #3

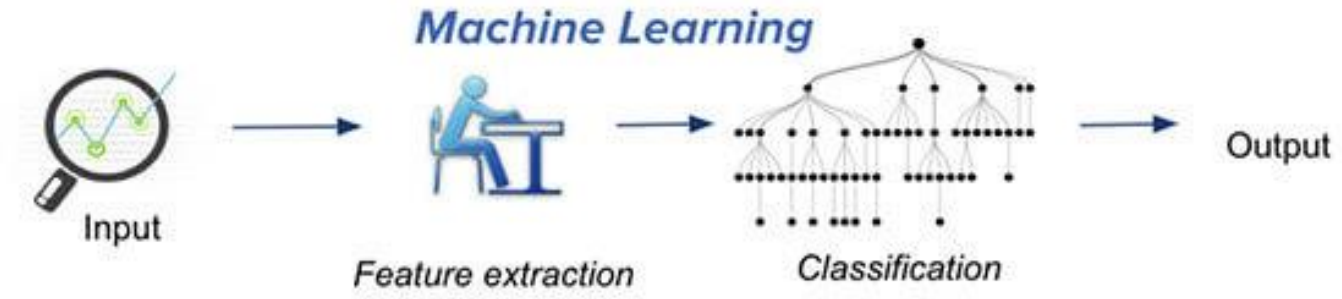


Reinforcement Learning

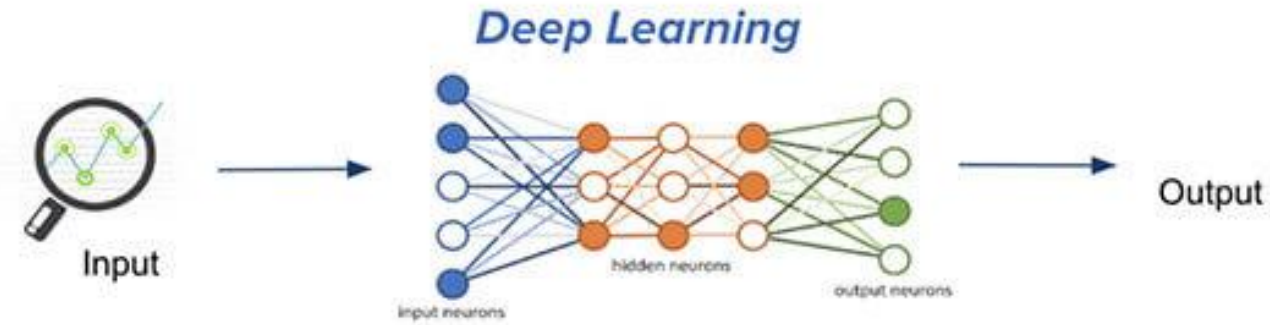
Multi-Agent Hide and Seek



What about *Deep Learning*?



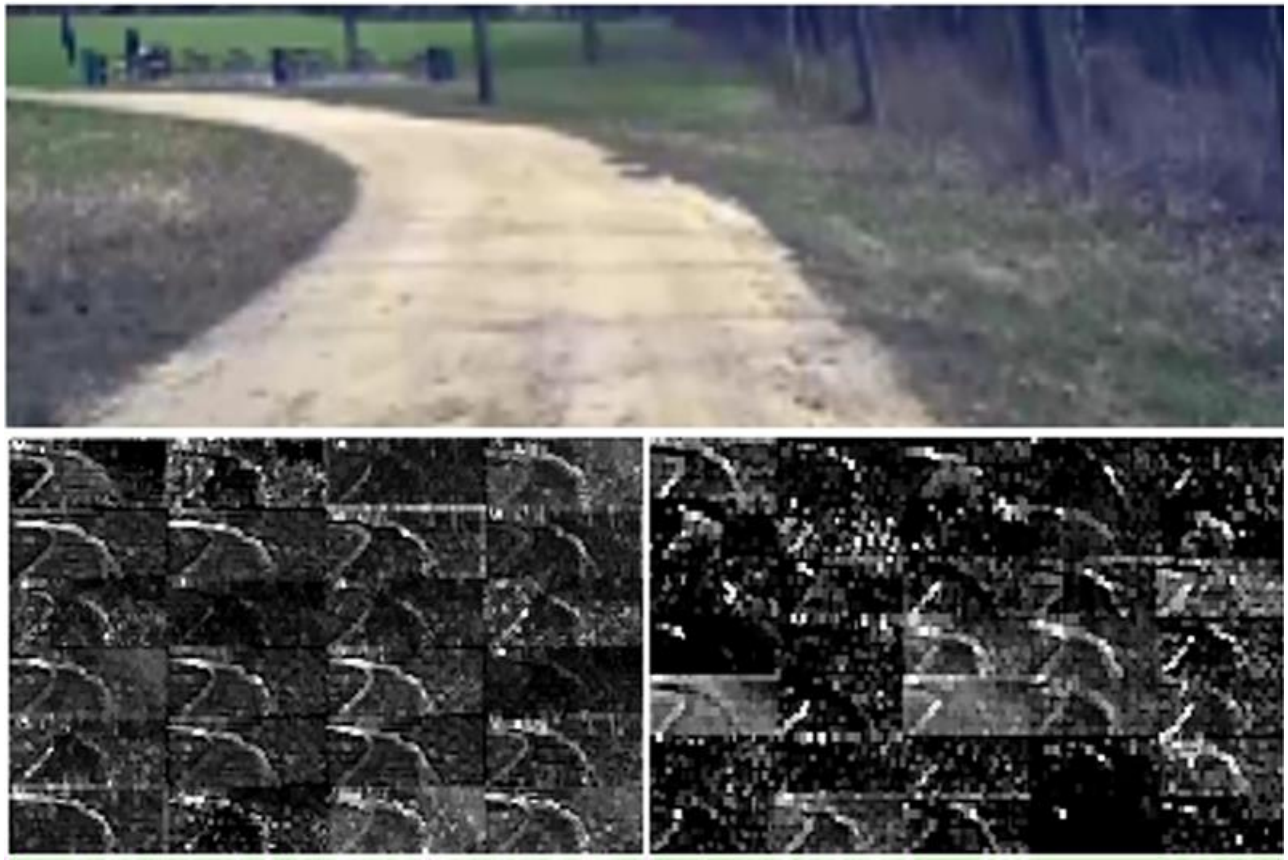
Traditional machine learning uses hand-crafted features, which is tedious and costly to develop.



Neural Networks

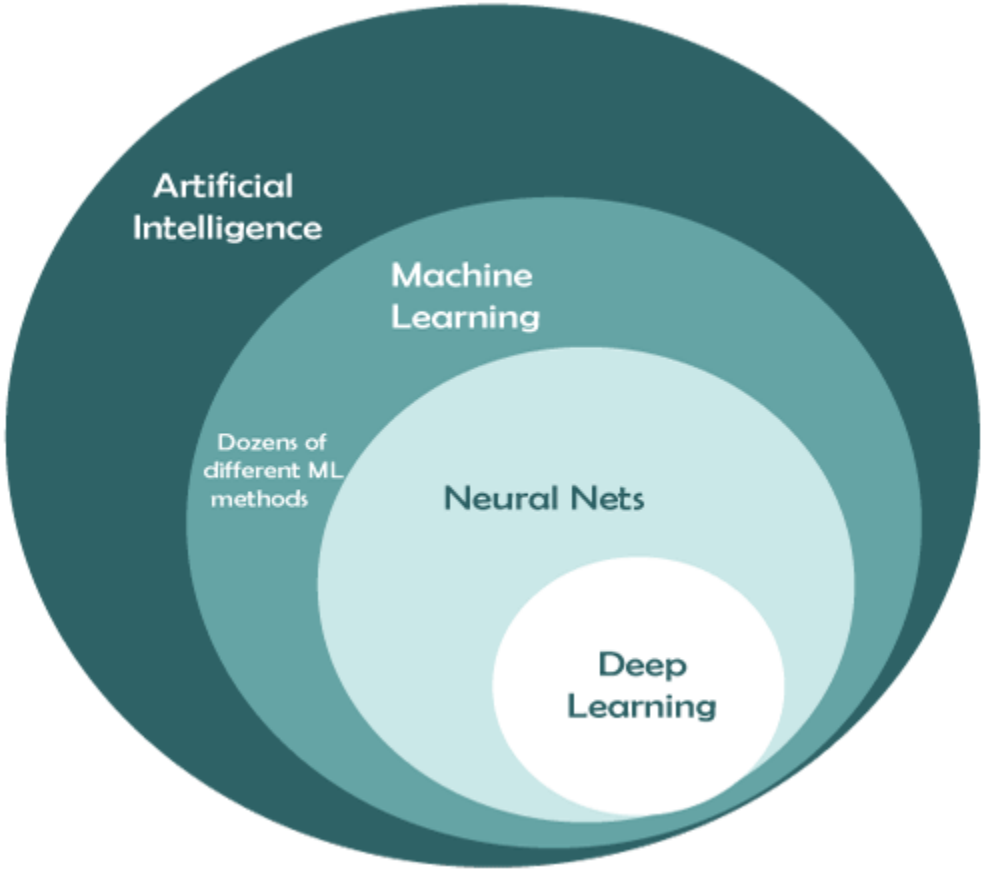
Deep learning learns hierarchical representation from the data itself, and scales with more data.

NVIDIA Self-Driving Car



How the CNN “sees” an unpaved road. Top: subset of the camera image sent to the CNN. Bottom left: Activation of the first layer feature maps. Bottom right: Activation of the second layer feature maps. This demonstrates that the CNN learned to detect useful road features on its own, i.e., with only the human steering angle as training signal. We never explicitly trained it to detect the outlines of roads.

AI vs *ML* vs *NN* vs *DL*



Classification

Dataset

	Attribute 1	Attribute 2	…	…	Attribute N	Class
X1						„Yes“
.						„No“
.						„Yes“
.						„Yes“
Xm						„Yes“

Xt = (a1, a2, … , aN)

Class(Xt) = ?

Classification

Dataset

	Humidity	Wind	Temperature	Class (Play Tennis)
X1	high	mid	35°	No
X2	mid	strong	5°	No
X3	mid	weak	20°	Yes
X4	high	weak	25°	Yes
X5	low	strong	18°	No

Xt = (low, weak, 21°)

Class(Xt) = ?

Regression

Dataset

	Attribute 1	Attribute 2	…	…	Attribute N	Class
X1						1.5
.						0.7
.						1.8
.						
Xm						1.2

Xt = (a1, a2, … , aN)

Class(Xt) = ?

Regression

Dataset

	Humidity	Wind	Outlook	Class (T°)
X1	high	mid	rainy	35
X2	mid	strong	overcast	5
X3	mid	weak	sunny	20
X4	high	weak	overcast	25
X5	low	strong	overcast	18

Xt = (low, weak, sunny)

Class(Xt) = ?

Clustering

Dataset

	Attribute 1	Attribute 2	…	…	Attribute N
X1					
.					
.					
.					
Xm					

$\forall\ xi \in D,\ xi = (a1, a2, \ldots, aN)$

$D = (x1, x2, \ldots, xm)$

Clusters (D) = ?

Clustering

Dataset

	Humidity	Wind	Temperature
X1	high	mid	35°
X2	mid	strong	5°
X3	mid	weak	20°
X4	high	weak	25°
X5	low	strong	18°

Data Preprocessing





Data Preprocessing



Data Preprocessing

- „Garbage in, garbage out“
- Data-gathering methods are often loosely controlled
  - Out-of-range values (e.g., Income: −100)
  - Impossible data combinations (e.g., Sex: Male, Pregnant: Yes)
  - Missing values
- Misleading results

Why Preprocessing ?

- Real world data are generally
  - Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
  - Noisy: containing errors or outliers
  - Inconsistent: containing discrepancies in codes or names
- Tasks in data preprocessing
  - Data cleaning: fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies.
  - Data integration: using multiple databases, data cubes, or files.
  - Data transformation: normalization and aggregation.
  - Data reduction: reducing the volume but producing the same or similar analytical results.
  - Data discretization: part of data reduction, replacing numerical attributes with nominal ones.

Data Cleaning

- Fill in missing values (attribute or class value):
  - Ignore the tuple: usually done when class label is missing.
  - Use the attribute mean (or majority nominal value) to fill in the missing value.
  - Use the attribute mean (or majority nominal value) for all samples belonging to the same class.
  - Predict the missing value by using a learning algorithm: consider the attribute with the missing value as a dependent (class) variable and run a learning algorithm (usually Bayes or decision tree) to predict the missing value.
- Identify outliers and smooth out noisy data:
  - Binning
    - Sort the attribute values and partition them into bins (see "Unsupervised discretization" below);
    - Then smooth by bin means, bin median, or bin boundaries.
  - Clustering: group values in clusters and then detect and remove outliers (automatic or manual)
  - Regression: smooth by fitting the data into regression functions.
- Correct inconsistent data: use domain knowledge or expert decision.

Data Transformation

- Normalization:
  - Scaling attribute values to fall within a specified range.
    - Example: to transform V in [min, max] to V' in [0,1], apply V'=(V-Min)/(Max-Min)
  - Scaling by using mean and standard deviation (useful when min and max are unknown or when there are outliers): V'=(V-Mean)/StDev
- Aggregation: moving up in the concept hierarchy on numeric attributes.
- Generalization: moving up in the concept hierarchy on nominal attributes.
- Attribute construction: replacing or adding new attributes inferred by existing attributes.

Data Reduction

- Reducing the number of attributes
  - Data cube aggregation: applying roll-up, slice or dice operations.
  - Removing irrelevant attributes: attribute selection (filtering and wrapper methods), searching the attribute space.
  - Principle component analysis (numeric attributes only): searching for a lower dimensional space that can best represent the data..
- Reducing the number of attribute values
  - Binning (histograms): reducing the number of attributes by grouping them into intervals (bins).
  - Clustering: grouping values in clusters.
  - Aggregation or generalization
- Reducing the number of tuples
  - Sampling

Data discretization

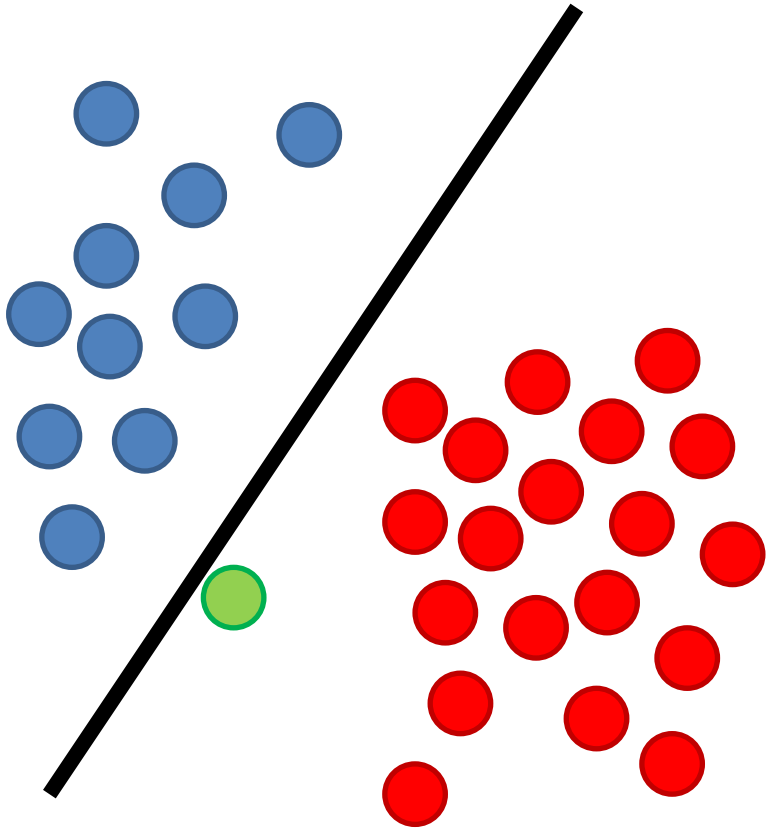
- Part of data reduction, replacing numerical attributes with nominal ones.
- Nominal to numerical?
- Binarization?

Classification Algorithms



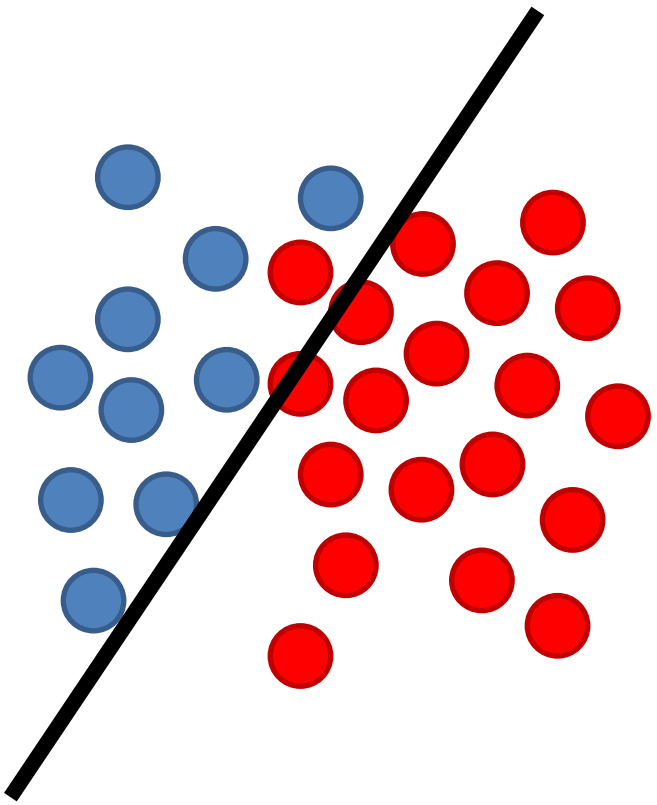
Classification Algorithms

- Linear classifiers
  - Logistic regression
  - Naive Bayes classifier
  - Perceptron



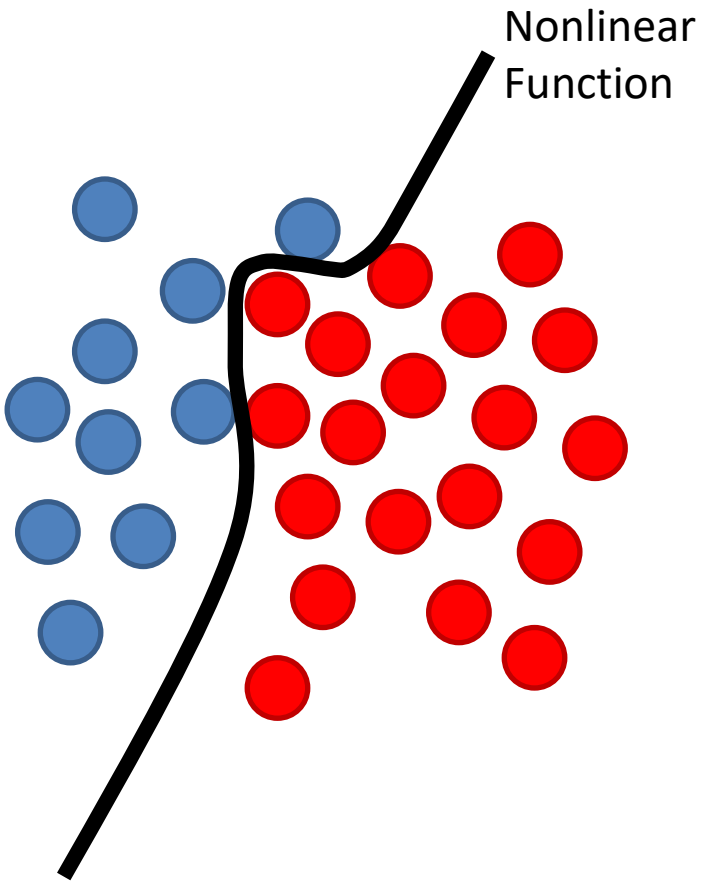
Classification Algorithms

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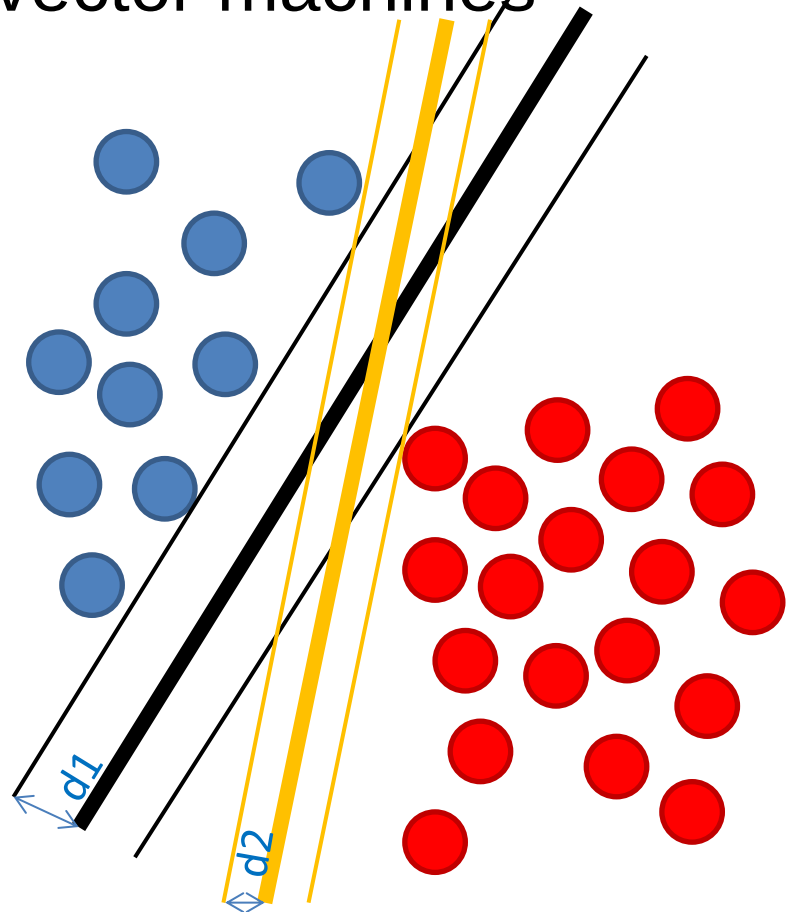
Classification Algorithms

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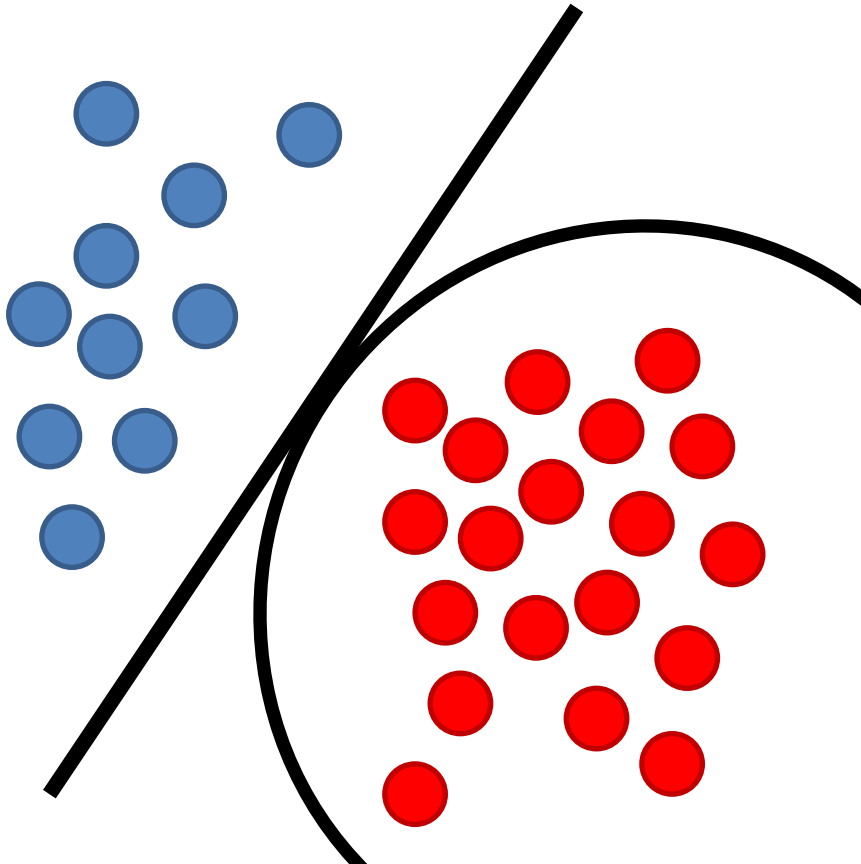
Classification Algorithms

- Support vector machines
  - Least squares support vector machines



Classification Algorithms

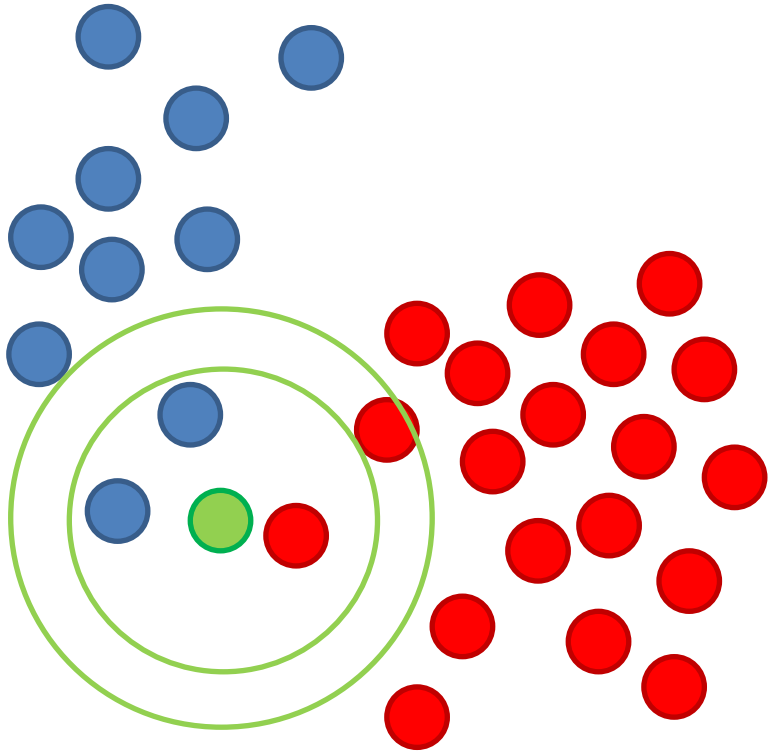
- Quadratic classifiers



Classification Algorithms

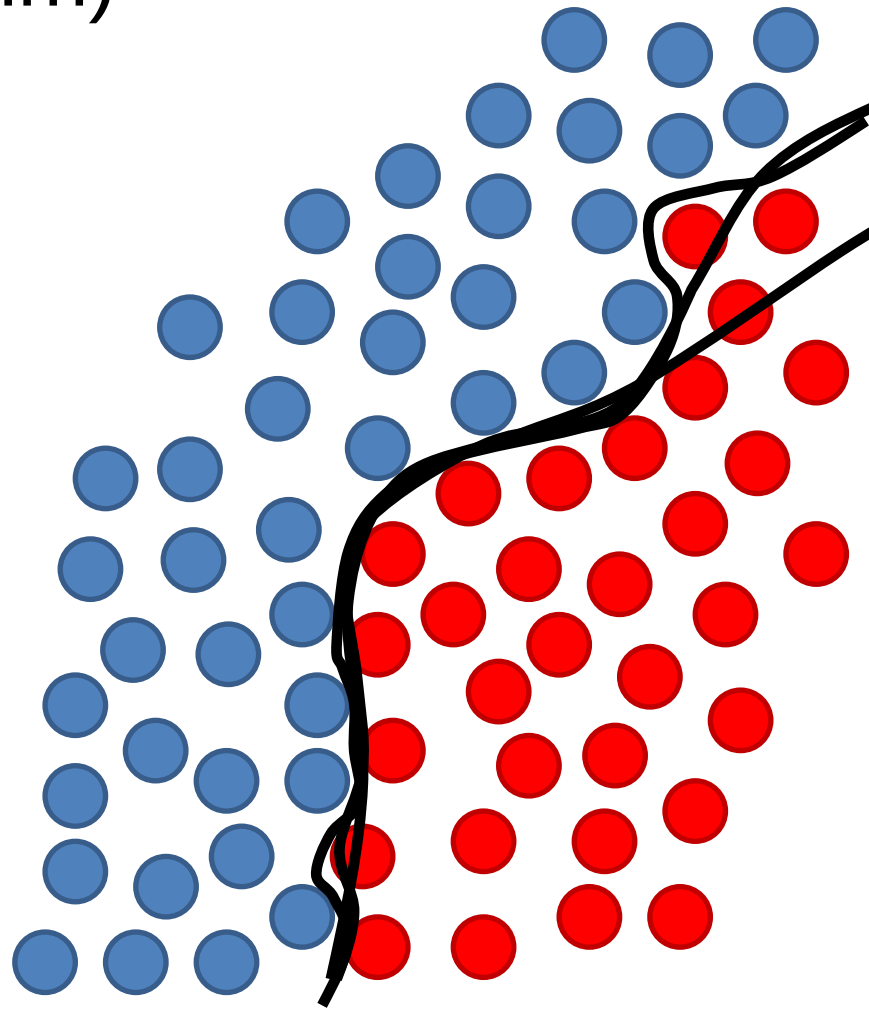
- Kernel estimation
  - k-nearest neighbor

K = 4



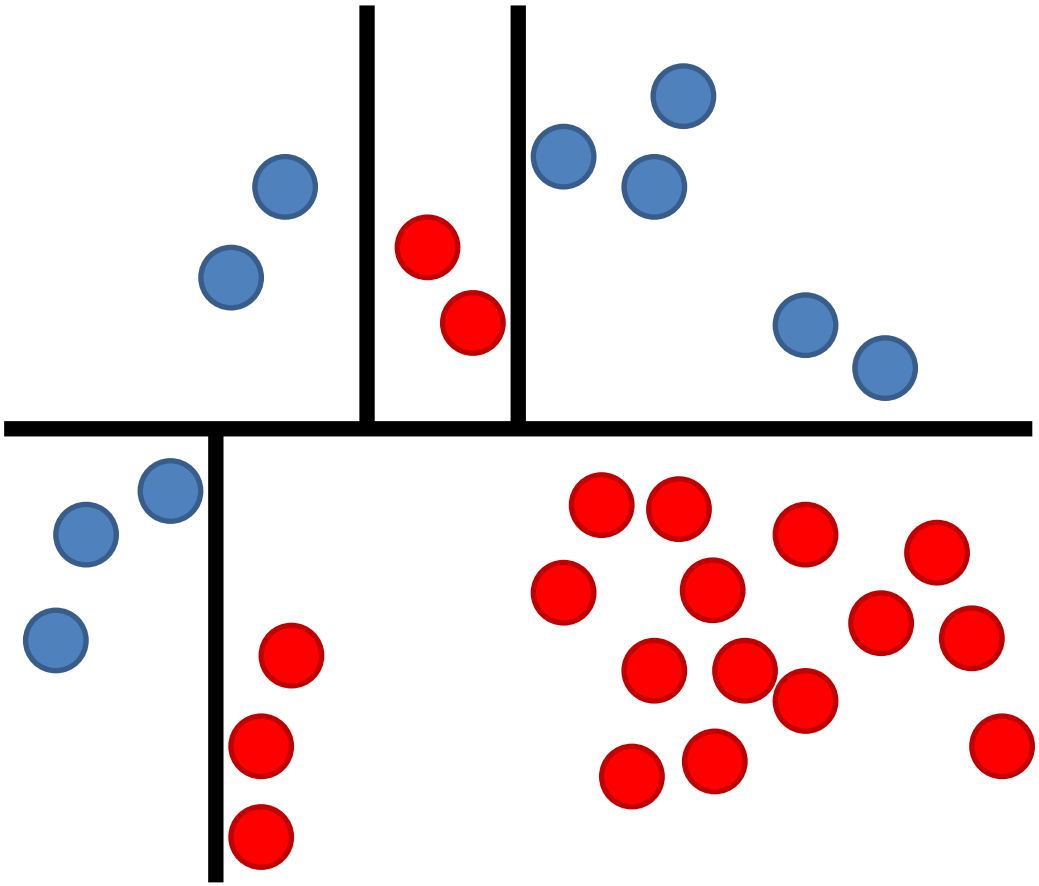
Classification Algorithms

- Boosting (meta-algorithm)
  - AdaBoost



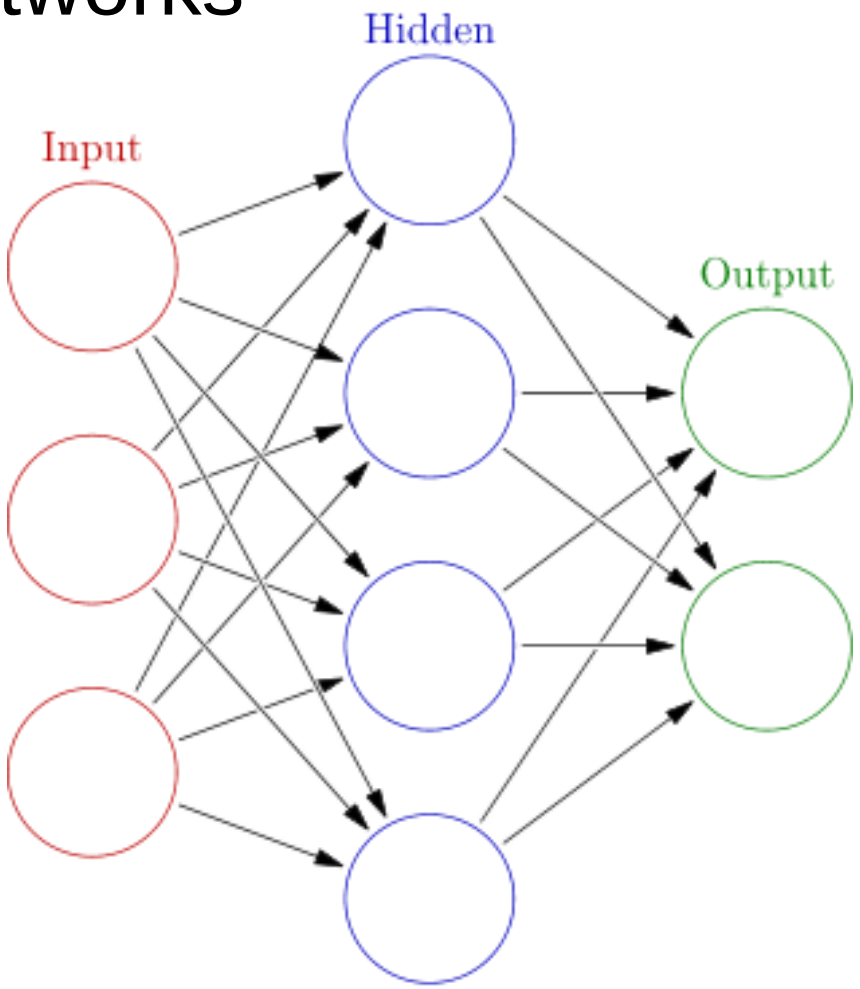
Classification Algorithms

- Decision trees
  - Random forests



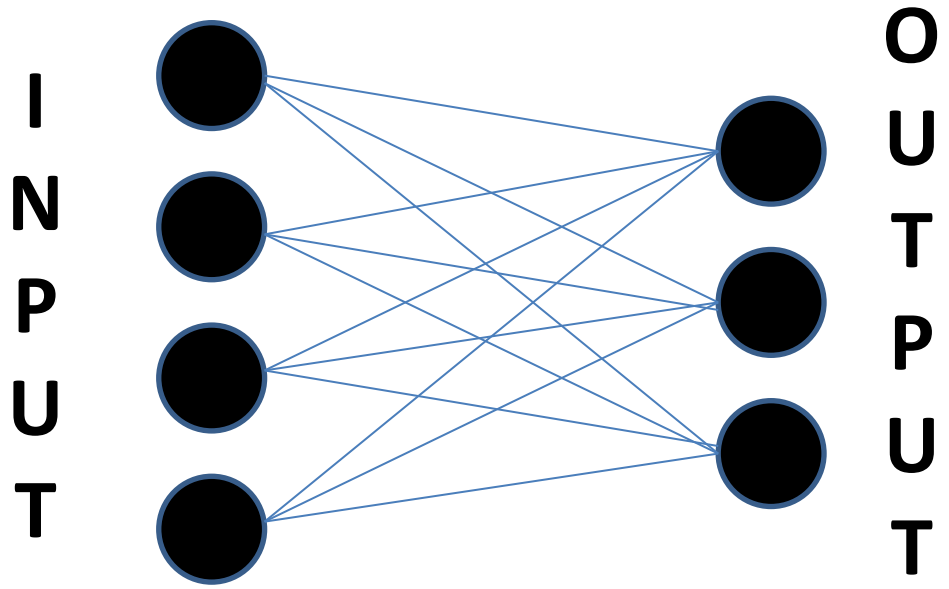
Classification Algorithms

- Neural networks



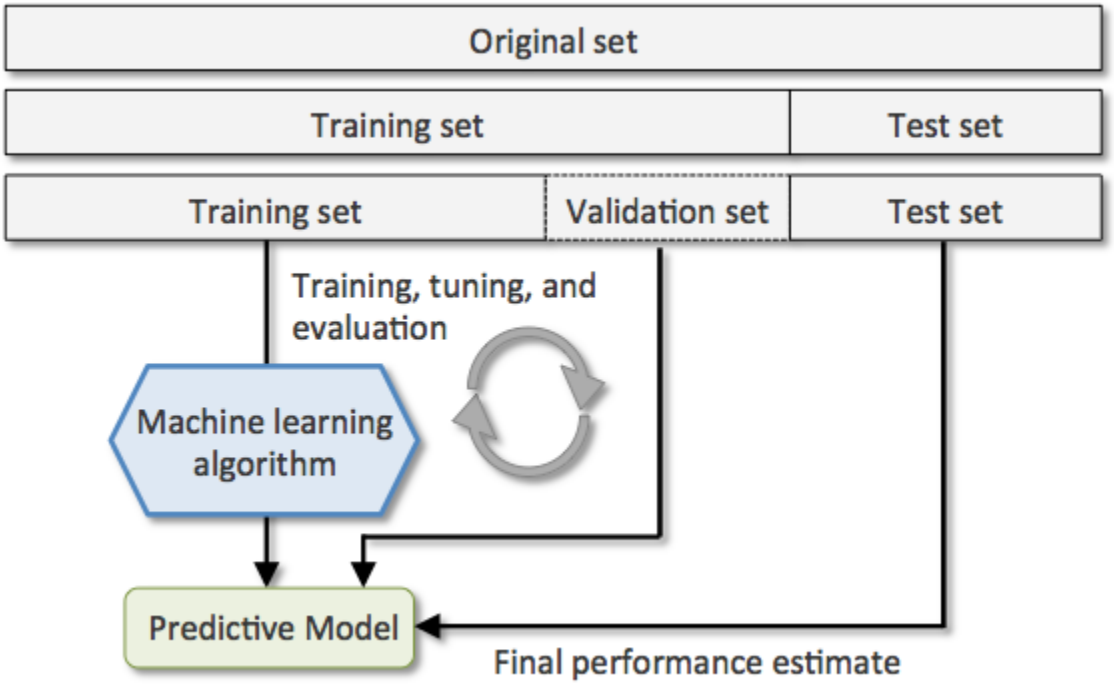
Classification Algorithms

- Learning vector quantization

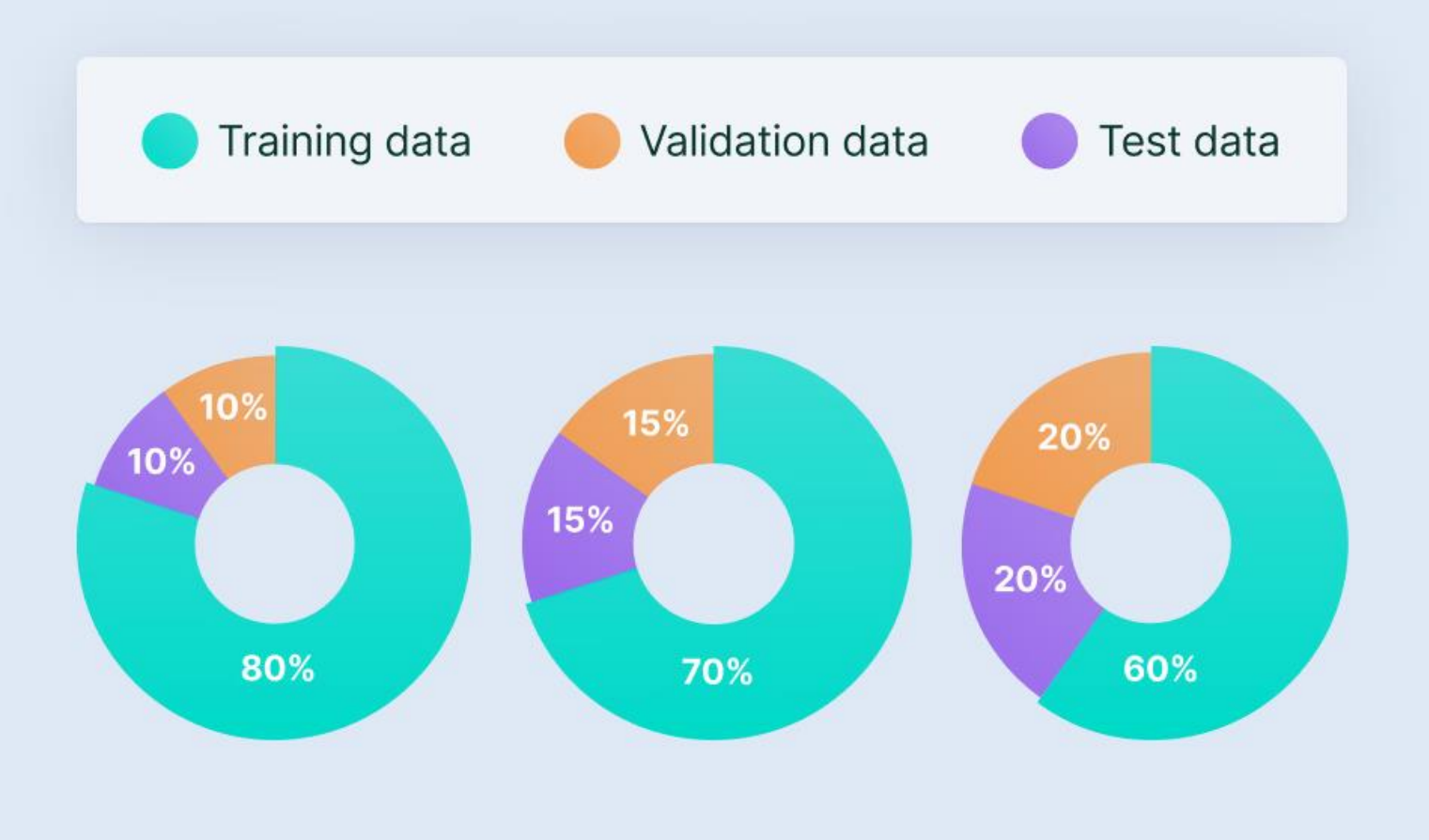


„winner-take-all“

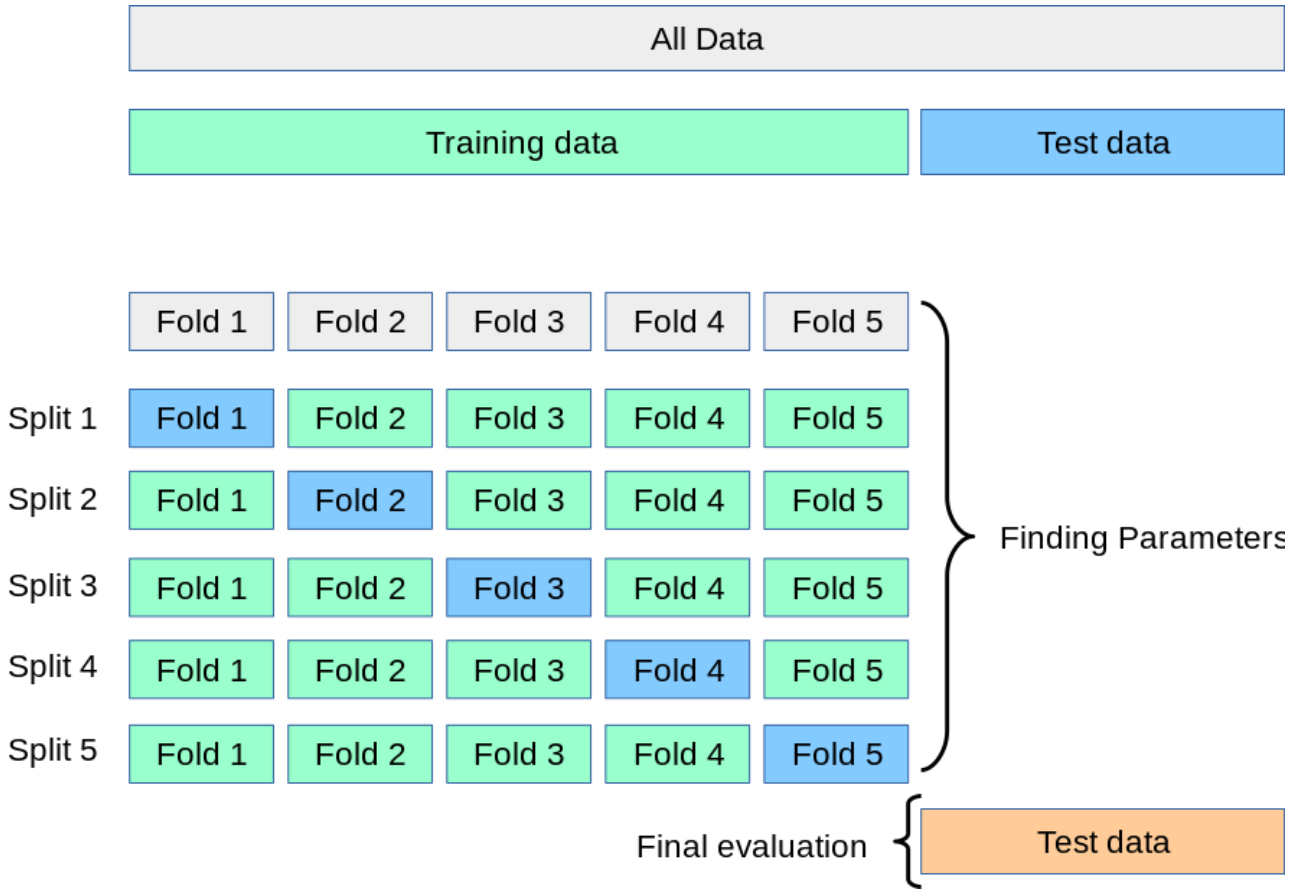
Train – Validation – Test



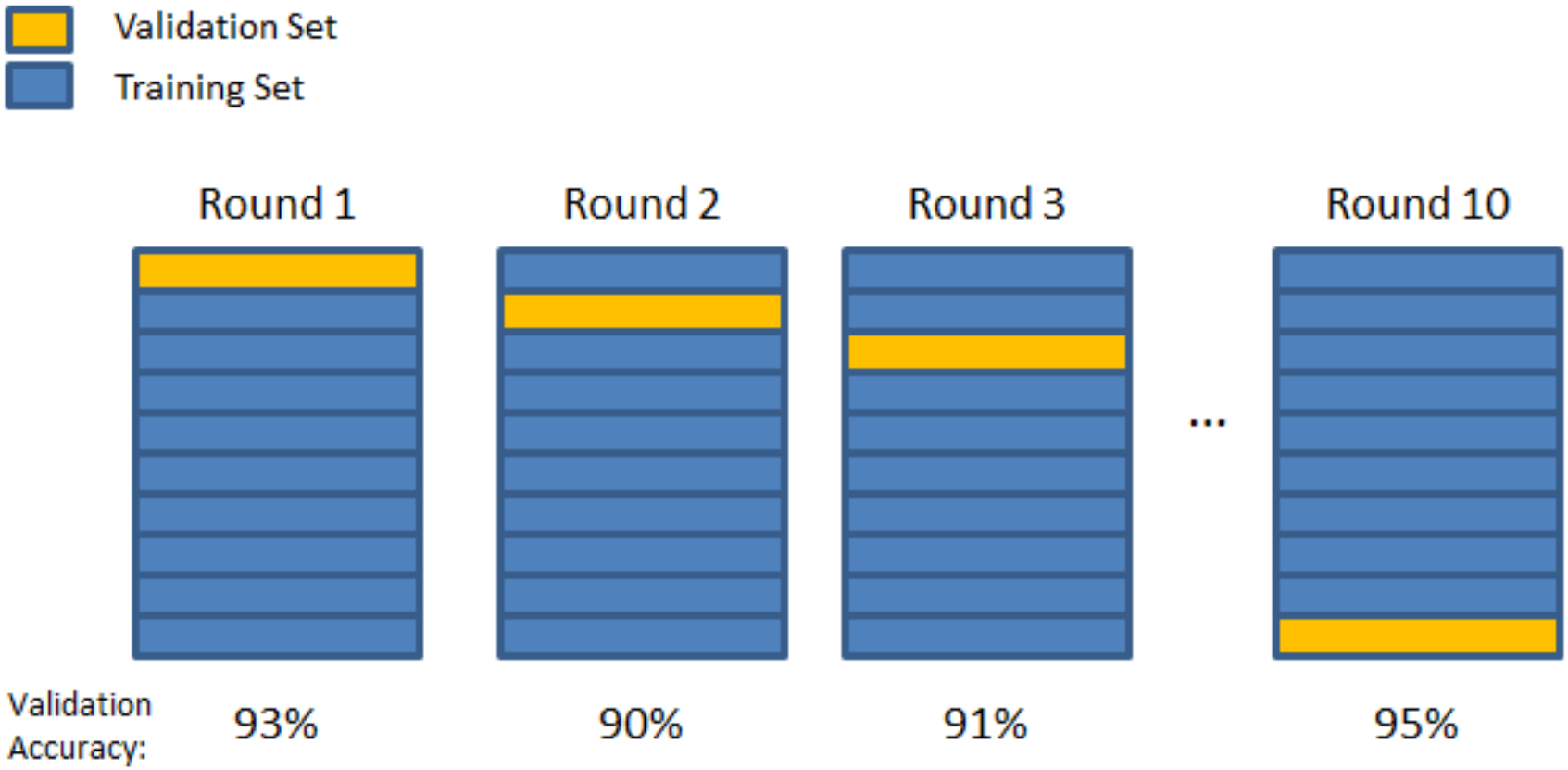
Train – Validation – Test



Cross-validation



k-Fold Cross-Validation

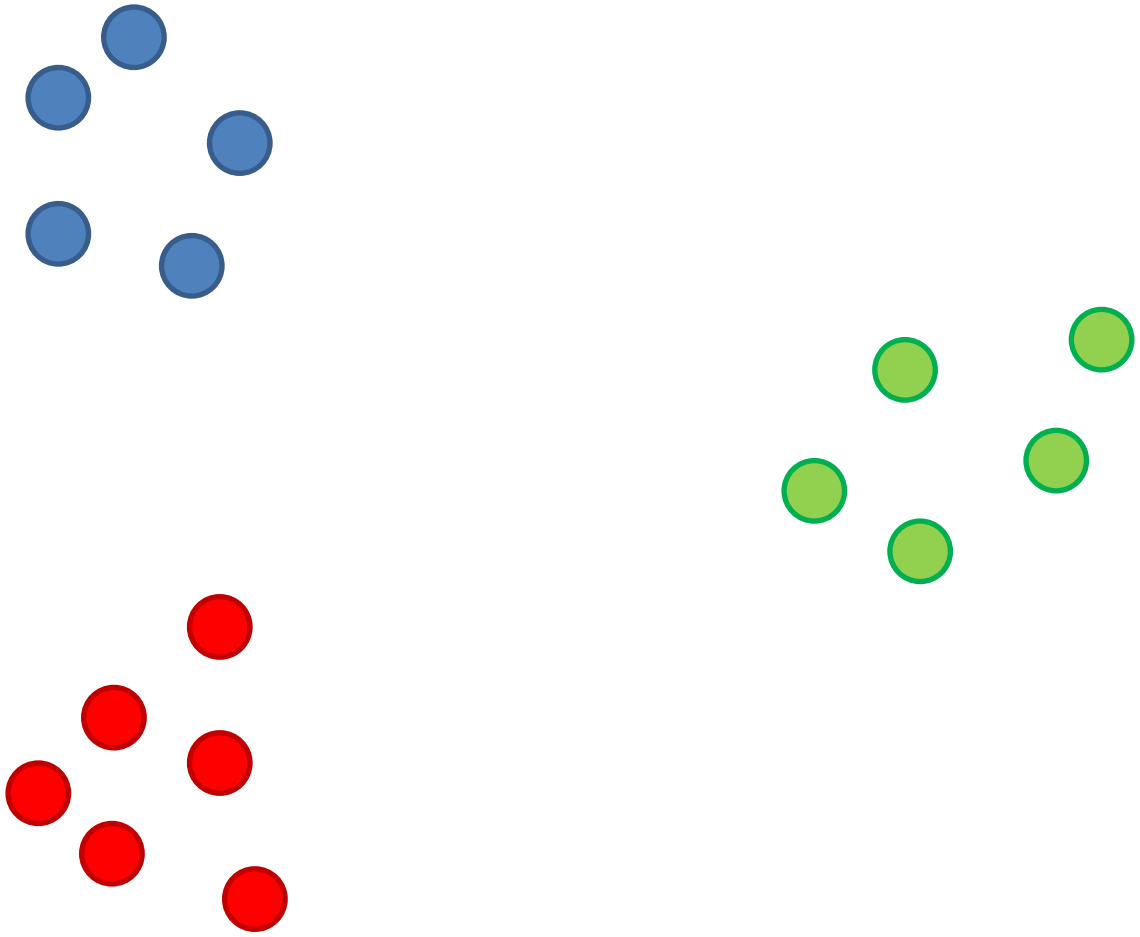


** The example is with 10-fold cross validation!*

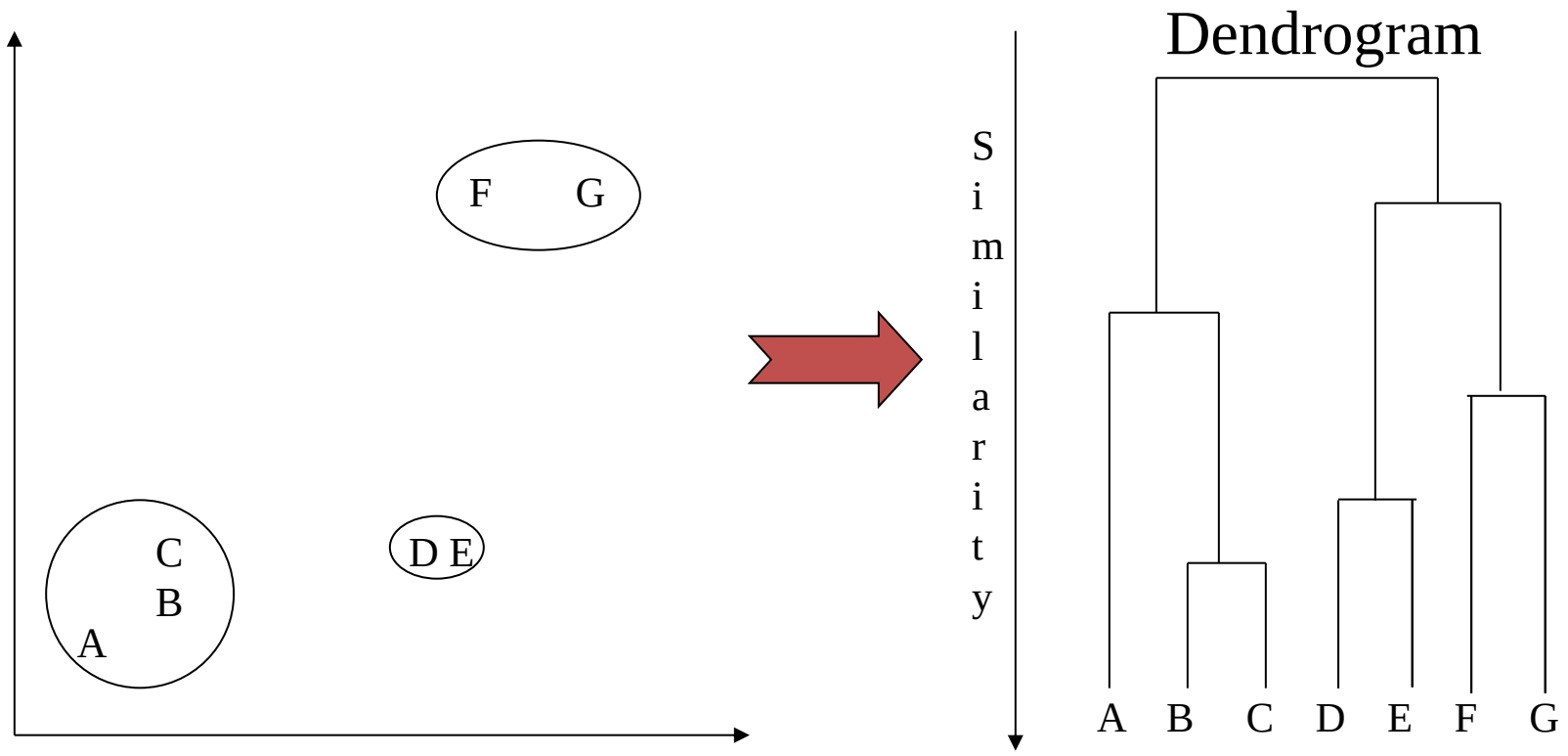
Clustering



Clustering Example

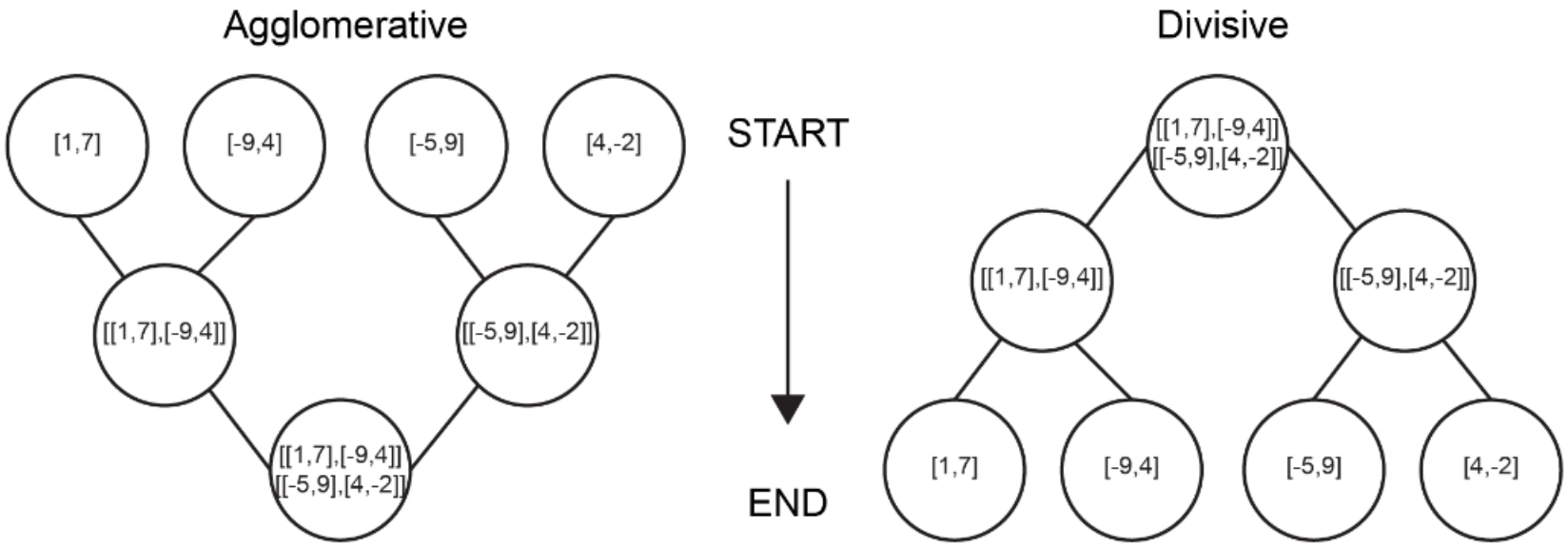


Hierarchical Clustering



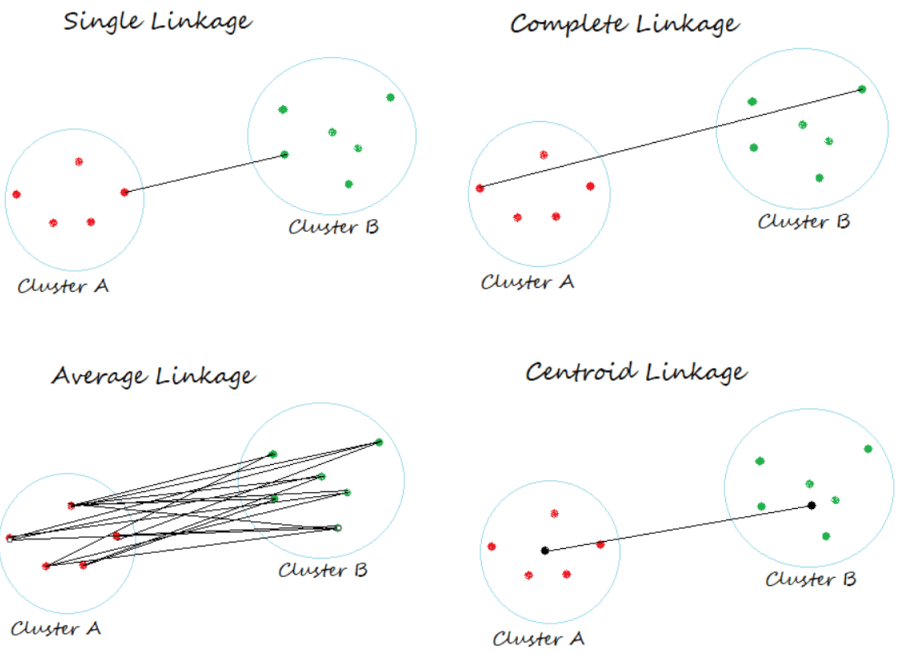
Hierarchical Clustering

- Agglomerative (bottom-up)
- Divisive (partitional, top-down)

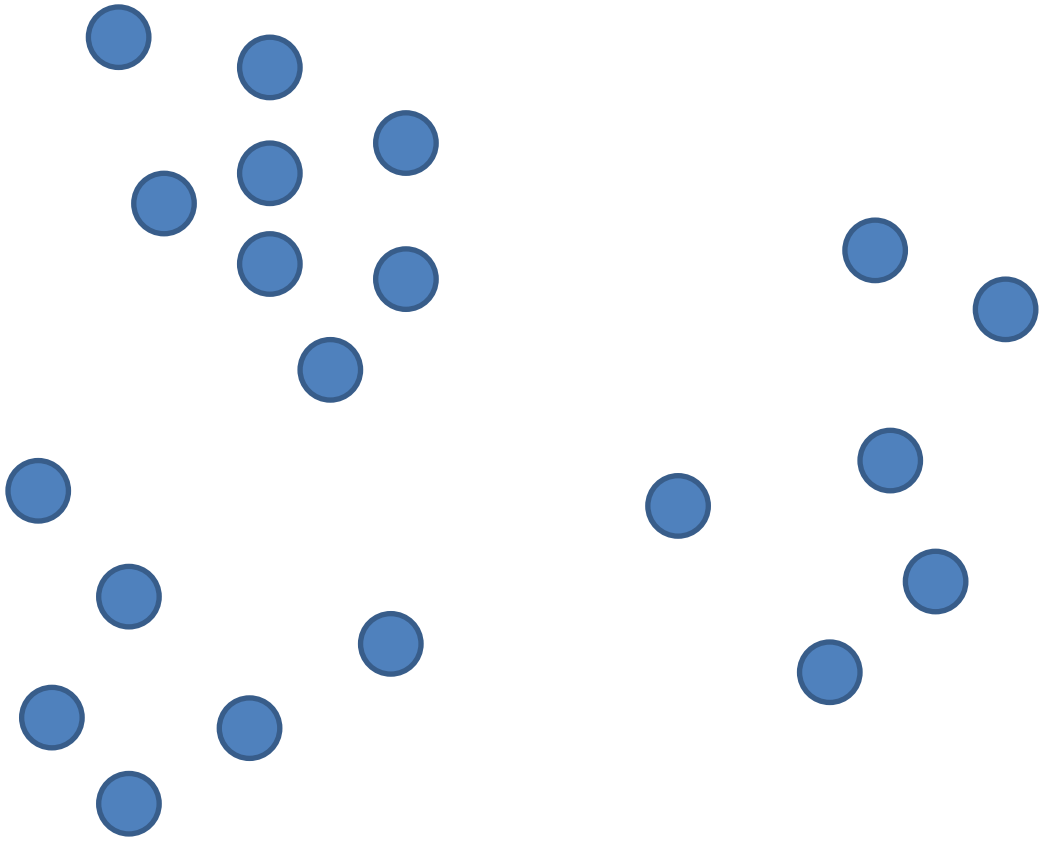


Cluster Similarity

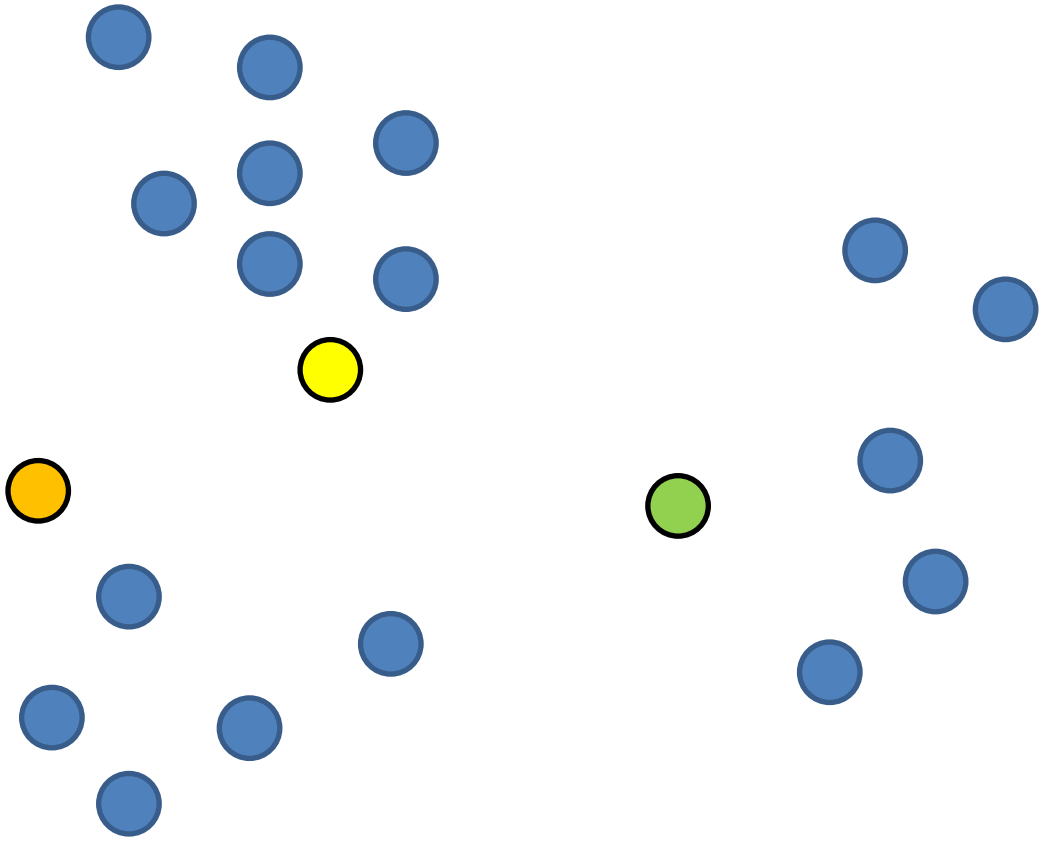
- **Single Link:** Similarity of the two most similar members.
- **Complete Link:** Similarity of the two least similar members.
- **Group Average:** Average similarity between members.



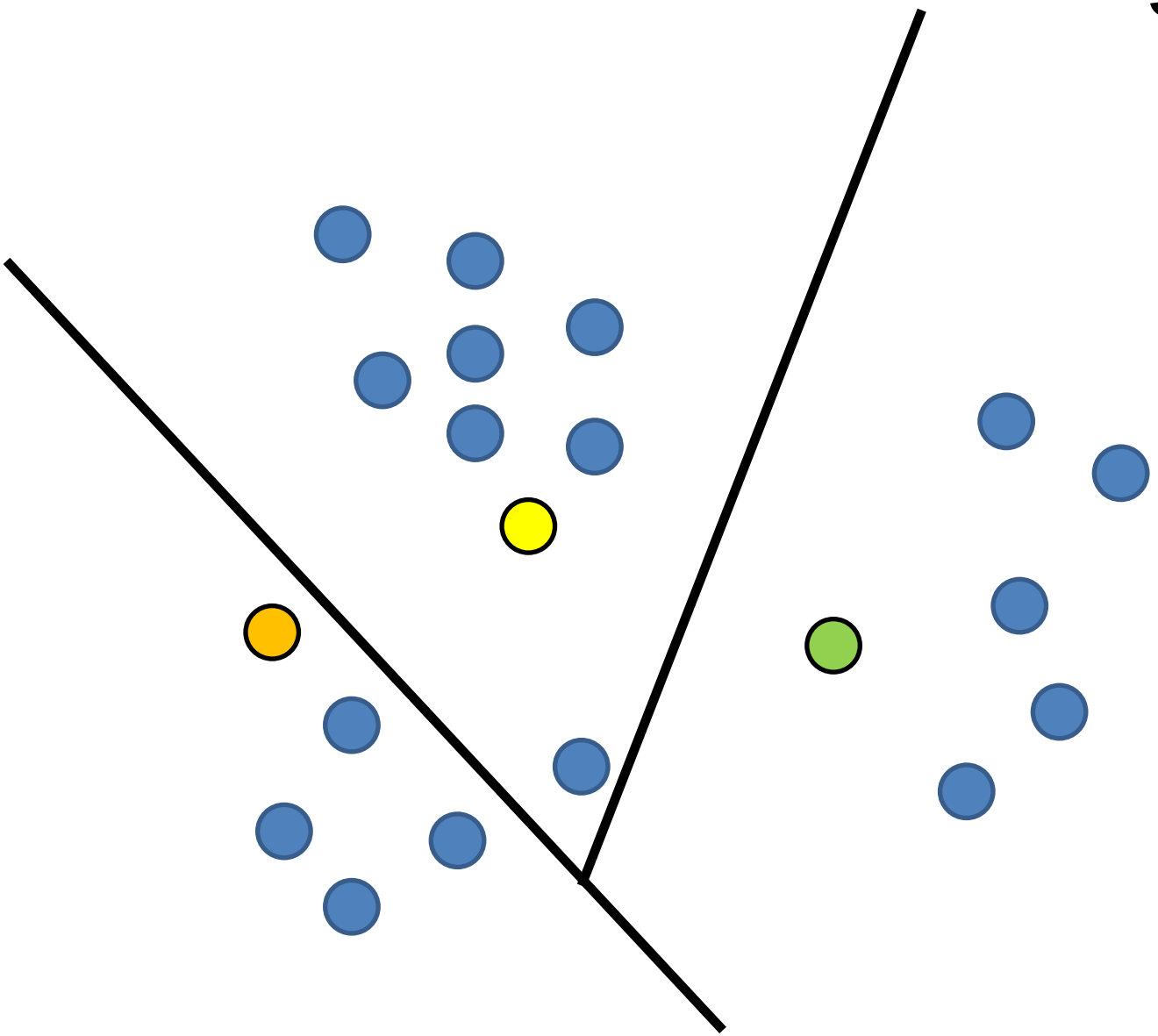
K-Means Clustering



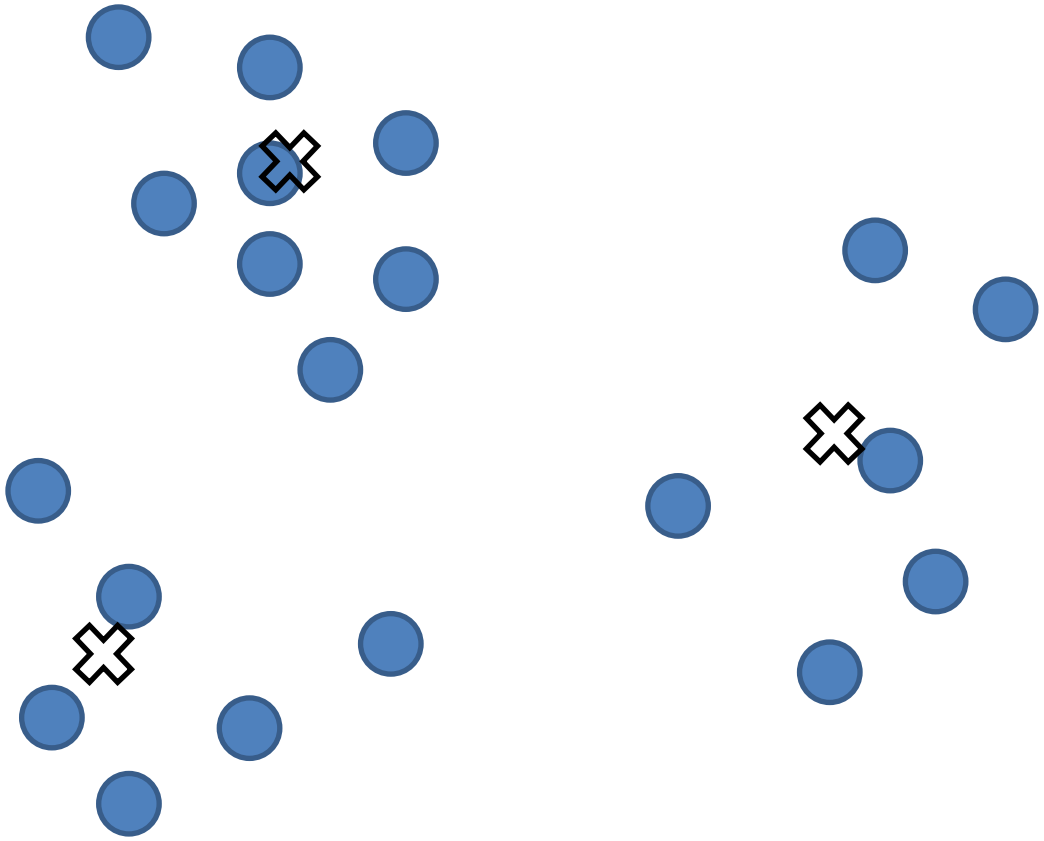
K-Means Clustering



K-Means Clustering



K-Means Clustering



K-Means Clustering

